

STATISTICS PAPER II · 2027 FORECAST

# Linear Models — 2027 Forecast

256 examiner-style forecast questions with exam shortcut and answer key

**256**

Forecast Questions

**21**

Topics Covered

## FORECAST · AI-GENERATED

No problem is copied from a past paper. Every explanatory note is an AI-derived shortcut, independently generated and verified by the builder of this platform, not an official answer key.

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# Matrix Theory for Linear Models

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6 forecast questions in this topic

Linear Models

Generalized Inverse of a Matrix

Medium

**Q1.**

If  $A$  is an  $m \times n$  matrix of rank  $r$ , and  $A^-$  is a generalized inverse satisfying  $AA^-A=A$ , what are the order of  $A^-$  and the minimum possible rank of  $A^-$  over all valid choices?

**A. Order  $n \times m$ ; minimum rank  $r$**

B. Order  $m \times n$ ; minimum rank  $r$

C. Order  $n \times m$ ; minimum rank  $\min(m,n)$

D. Order  $n \times m$ ; minimum rank  $m$

**Correct answer: A**

## EXAM SHORTCUT

Conformability forces  $A^-$  to be  $n \times m$ ; a reflexive g-inverse achieves the lower bound  $\text{rank}(A^-)=r$ .

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Linear Models

Generalized Inverse of a Matrix

Medium

**Q2.**

For a symmetric matrix  $A$ , which statement about its generalized inverse is correct?

**A. A symmetric generalized inverse can always be found, but not every g-inverse of a symmetric matrix is symmetric**

B. Symmetric matrices never possess a generalized inverse

C. A g-inverse of a symmetric matrix is symmetric only if  $A$  is also idempotent

D. A generalized inverse of a symmetric matrix is always itself symmetric

**Correct answer: A**

## EXAM SHORTCUT

Spectral decomposition  $A=PD\bar{P}'$  gives a ready-made symmetric g-inverse  $PD\bar{P}'$ , but the g-inverse family is much larger and includes non-symmetric members too.

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Q3.

Let  $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ . Which of the following matrices  $G$  satisfies  $AGA=A$  (i.e. is a generalized inverse of  $A$ )?

A.  $\begin{pmatrix} 0.5 & 0 \\ 0 & 0.5 \end{pmatrix}$

B.  $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

C.  $\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$

D. ✓  $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$

Correct answer: D

## EXAM SHORTCUT

$A$  has rank 1; test each  $G$  by computing  $AGA$  and checking it reproduces  $A$  exactly - only option (d) survives.

Q4.

For the hat matrix  $H=X(X'X)^{-1}X'$  in a full-rank regression model, which statement is FALSE?

A.  $\text{Cov}(e_i, e_j) = -\sigma^2 h_{ij}$  for  $i \neq j$

B. ✓  $\text{Cov}(e_i, e_j) = +\sigma^2 h_{ij}$  for  $i \neq j$

C.  $\sum_i h_{ii} = p$ , the number of parameters

D.  $0 \leq h_{ii} \leq 1$  for every  $i$

Correct answer: B

## EXAM SHORTCUT

Residual covariance carries a MINUS sign:  $\text{Cov}(e_i, e_j) = -\sigma^2 h_{ij}$ ; the '+' version is the planted false statement.

Q5.

If  $P$  is an idempotent matrix ( $P^2=P$ ) of order  $n \times n$  with rank  $r$ , its eigenvalues are

A. All equal to 1

B. All equal to  $r/n$

C. ✓ All 0 or 1, with exactly  $r$  eigenvalues equal to 1

D. All equal to  $\pm 1$

Correct answer: C

## EXAM SHORTCUT

Idempotency forces eigenvalues to satisfy  $\lambda^2 = \lambda$ , so  $\lambda \in \{0, 1\}$ ; the count of 1's equals  $\text{tr}(P) = \text{rank}(P) = r$ .

**Q6.**

If  $Y \sim N_n(X\beta, \sigma^2 I)$  and  $H = X(X'X)^{-1}X'$  has rank  $p$ , then  $(Y - \hat{X}\hat{\beta})'(Y - \hat{X}\hat{\beta})/\sigma^2$  follows

- A.  $\chi^2$  with  $p$  degrees of freedom
- B. an  $F$  distribution with  $(p, n-p)$  degrees of freedom
- C.  ~~$\chi^2$~~  with  $(n-p)$  degrees of freedom
- D.  $\chi^2$  with  $n$  degrees of freedom

**Correct answer: C****EXAM SHORTCUT**

Residual SS  $= e'e = Y'(I-H)Y$  with  $(I-H)$  idempotent of rank  $n-p$ , giving  $\chi^2_{n-p}$  under normality.

# Theory of Linear Estimation

13 forecast questions in this topic

Linear Models

Gauss-Markov Theorem, Normal Equations and BLUE

Hard

Q1.

Let  $X_i, Y_i, Z_i$  ( $i=1,2,3$ ) be nine independent observations with common variance  $\sigma^2$ , and  $E(X_i)=\theta_1+\theta_2$ ,  $E(Y_i)=\theta_1-\theta_2$ ,  $E(Z_i)=2\theta_1$ . Writing  $X, Y, Z$  for  $\sum_{i=1}^3 X_i, \sum_{i=1}^3 Y_i, \sum_{i=1}^3 Z_i$ , the BLUE of  $\theta_1$  is

A.  $\frac{1}{18}(2X+2Y+Z)$

B.  $\frac{1}{18}(X+Y+2Z)$

C.  $\frac{1}{9}(X+Y+Z)$

D.  $\frac{1}{18}(X-Y+2Z)$

Correct answer: B

## EXAM SHORTCUT

$X \sim 3(\theta_1 + \theta_2)$ ,  $Y \sim 3(\theta_1 - \theta_2)$ ,  $Z \sim 6\theta_1$ , each with variance  $3\sigma^2$  and mutually independent; minimize  $a^2 + b^2 + c^2$  subject to unbiasedness for  $\theta_1$ .

Linear Models

Regression Specifics, Design Matrix, Multicollinearity

Medium

Q2.

For the model  $y_i = \beta_1 + \beta_2(x_i - \alpha)$ ,  $i=1, \dots, n$ , the least-squares estimators  $\hat{\beta}_1$  and  $\hat{\beta}_2$  are uncorrelated when  $\alpha$  equals

A.  $\sum x_i^2/n$

B.  $\bar{x}$  (the sample mean of  $x$ )

C. the sample median of  $x$

D. 0

Correct answer: B

## EXAM SHORTCUT

$X'X$  is diagonal exactly when  $\sum (x_i - \alpha) = 0$ , i.e.  $\alpha = \bar{x}$  - the standard centering trick.

Q3.

In the Gauss-Markov model  $Y=X\beta+e$ , a linear parametric function  $\lambda'\beta$  is estimable if and only if

- A.  $X$  has full column rank
- B.  $\lambda'\lambda=1$
- C.  $\lambda$  lies in the null space of  $X$
- D.  $\lambda'=a'X$  for some vector  $a$  (i.e.  $\lambda$  lies in the row space of  $X$ )

Correct answer: D

#### EXAM SHORTCUT

Estimability is exactly membership of  $\lambda'$  in the row space of  $X$  (equivalently, of  $X'X$ ).

Q4.

For the model  $Y=X\beta+e$  with possibly singular  $X'X$ , if  $G$  is any generalized inverse of  $X'X$ , then  $XGX'X$  equals

- A. the identity matrix
- B. the zero matrix
- C.  $X'X$
- D.  $X$ , for every choice of generalized inverse  $G$

Correct answer: D

#### EXAM SHORTCUT

$XGX'X=X$  for any g-inverse  $G$  of  $X'X$  - the key lemma behind invariance of fitted values  $\hat{Y}=X\hat{\beta}$  to the choice of  $G$ .

Q5.

In simple linear regression with  $n=5$ ,  $\bar{x}=3$ ,  $\sum (x_i-\bar{x})^2=20$  and error variance  $\sigma^2=4$ ,  $\text{Cov}(\hat{\beta}_0, \hat{\beta}_1)$  equals

- A. 0.6
- B.  $-0.6$
- C. 0
- D. -1.2

Correct answer: B

#### EXAM SHORTCUT

$\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\sigma^2 \bar{x} / \sum (x_i - \bar{x})^2 = -4(3)/20 = -0.6$ .

Q6.

For nested linear models where Model 1 (with  $p_1$  parameters) is a special case of Model 2 (with  $p_2 > p_1$  parameters), both fitted to the same data by least squares, which is necessarily true of the residual sums of squares?

- A.  $RSS_1 = RSS_2$  always
- B.  $RSS_1 < RSS_2$  always
- C.  ~~$RSS_1 \geq RSS_2$  always, and  $RSS_1 - RSS_2 = 0$  is possible even when  $p_2 > p_1$~~
- D.  $RSS_1 - RSS_2$  can be negative

Correct answer: C

## EXAM SHORTCUT

Model 2's larger parameter space contains Model 1's fit as a special case, so its minimized RSS can only be  $\leq$  Model 1's.

Q7.

In the model  $Y = X\beta + e$  with  $D(e) = \sigma^2 G$  ( $G$  known, positive definite), transforming  $Z = G^{-1/2}Y$  reduces the model to Gauss-Markov form. The resulting GLS estimator of  $\beta$  is

- A.  $(X'X)^{-1}X'Y$
- B.  $(X'GX)^{-1}X'GY$
- C.  ~~$(X'G^{-1}X)^{-1}X'G^{-1}Y$~~
- D.  $(X'G^2X)^{-1}X'G^2Y$

Correct answer: C

## EXAM SHORTCUT

OLS on the transformed model  $Z = WY$ ,  $W = G^{-1/2}X$  gives  $(W'W)^{-1}W'Z = (X'G^{-1}X)^{-1}X'G^{-1}Y$  since  $G^{-1/2}'G^{-1/2} = G^{-1}$ .

Q8.

In the heteroscedastic model  $y_i = \beta x_i + e_i$  (no intercept) with  $\text{Var}(e_i) = \sigma^2/w_i$  for known weights  $w_i$ , the weighted least squares estimator of  $\beta$  is

A.  $\frac{\sum w_i x_i y_i}{\sum w_i x_i^2}$

B.  $\frac{\sum y_i}{\sum x_i}$

C.  $\frac{\sum x_i y_i}{\sum x_i^2}$

D.  $\frac{\sum w_i y_i}{\sum w_i x_i}$

Correct answer: A

#### EXAM SHORTCUT

Minimizing  $\sum w_i (y_i - \beta x_i)^2$  and setting the derivative to zero gives  $\hat{\beta} = \sum w_i x_i y_i / \sum w_i x_i^2$ .

Q9.

For the design matrix  $X$  ( $n \times p$ ) in a linear model,  $\text{rank}(X'X)$  equals

A.  ~~$\text{rank}(X)$~~

B.  $\min(n, p)$ , regardless of  $X$

C.  $p$ , always

D.  $n$ , always

Correct answer: A

#### EXAM SHORTCUT

$\text{rank}(X'X) = \text{rank}(X)$  is a standard linear-algebra identity, true whether or not  $X$  has full column rank.

Q10.

When  $X$  has full column rank  $p$ , the Gauss-Markov (BLUE) estimator of  $\beta$  itself is

A. non-unique even though  $X'X$  is invertible

B. unique only for estimable functions, not for  $\beta$  itself

C. ~~unique, equal to  $(X'X)^{-1}X'Y$~~

D. undefined

Correct answer: C

#### EXAM SHORTCUT

Full column rank makes  $X'X$  invertible, so every  $\lambda'\beta$  (including  $\beta$  itself) is estimable with a unique BLUE  $(X'X)^{-1}X'Y$ .

**Q11.**

In a simple linear regression with error  $df = 10$ ,  $\hat{\beta}_1 = 2.5$ ,  $SE(\hat{\beta}_1) = 0.5$ , and  $t_{0.025, 10} = 2.228$ , a 95% confidence interval for  $\beta_1$  is

**A. ~~1.39, 3.61~~**

B. (0.50, 4.50)

C. (2.00, 3.00)

D. (1.50, 3.50)

**Correct answer: A****EXAM SHORTCUT** $2.5 \pm 2.228(0.5) = 2.5 \pm 1.114 = (1.386, 3.614).$ **Q12.**

Under severe multicollinearity among the regressors (with  $X$  still of full column rank), the OLS estimator  $\hat{\beta} = (X'X)^{-1}X'Y$  remains

A. completely unaffected

**B. ~~unbiased, but with inflated variance and unstable estimates~~**

C. no longer a linear estimator

D. biased and inconsistent

**Correct answer: B****EXAM SHORTCUT**

Gauss-Markov only needs full rank, not a well-conditioned  $X'X$ ; multicollinearity inflates  $(X'X)^{-1}$ 's diagonal (variance) without breaking unbiasedness.

**Q13.**

The Gauss-Markov theorem states that, under  $E(e) = 0$ ,  $D(e) = \sigma^2 I$ , the least-squares estimator of any estimable  $\lambda'\beta$  is

A. the MLE, which requires a normality assumption

**B. ~~BLUE among all linear unbiased estimators, without requiring normality~~**

C. unbiased only, with no minimum-variance guarantee

D. the UMVUE among ALL unbiased estimators, not just linear ones

**Correct answer: B****EXAM SHORTCUT**

Gauss-Markov's guarantee is BLUE - best among LINEAR unbiased estimators - and needs no distributional assumption at all.

# Analysis of Variance

6 forecast questions in this topic

Linear Models

Two-way Classification and ANOVA

Easy

**Q1.**

In a two-way classification with interaction, factor  $A$  at 3 levels, factor  $B$  at 4 levels, and 5 observations per cell, the error degrees of freedom equal

A. 12

B. 44

**C. 48**

D. 59

**Correct answer: C**

## EXAM SHORTCUT

Error df  $= ab(m-1) = 3 \times 4 \times 4 = 48$ .

Linear Models

One-way Classification and ANOVA

Easy

**Q2.**

In a one-way ANOVA with 4 treatments and 5 observations per treatment, Treatment SS = 45 and Total SS = 105. The value of the  $F$  statistic is

A. 5.00

B. 6.00

C. 3.00

**D. 4.00**

**Correct answer: D**

## EXAM SHORTCUT

Error SS  $= 105 - 45 = 60$ ;  $F = (45/3)/(60/16) = 15/3.75 = 4$ .

Linear Models

Fixed, Random and Mixed Effects Models

Hard

**Q3.**

In a two-way ANOVA with both factors random and replication present, testing  $H_0$ : no main effect of factor  $A$  requires the  $F$ -test denominator to be

A. MSE, the error mean square

B. MS(B)

C. the total mean square

**D. MS(AB), the interaction mean square**

**Correct answer: D**

## EXAM SHORTCUT

Under the random-effects model,  $E[MS(A)] = \sigma_e^2 + m\sigma_{AB}^2 + b\sigma_A^2$  and  $E[MS(AB)] = \sigma_e^2 + m\sigma_{AB}^2$ , so  $F = MS(A)/MS(AB)$  isolates  $\sigma_A^2$ .

**Q4.**

Which multiple comparison procedure uses the studentized range distribution and a SINGLE fixed critical value for all pairwise comparisons among treatment means (with equal sample sizes)?

- A. Bonferroni's method
- B. Student-Newman-Keuls (SNK) test
- C. ~~Tukey's~~ **HSD test**
- D. Scheffe's test

**Correct answer: C****EXAM SHORTCUT**

Tukey's HSD: one studentized-range critical value applied uniformly to all pairwise differences - the most standard equal- $n$  pairwise procedure.

**Q5.**

In one-way ANOVA testing equality of  $k$  treatment means from  $N$  total observations, the statistic  $F = \text{MST}/\text{MSE}$  follows, under  $H_0$ , the distribution

- A.  $F_{N-1, N-k}$
- B.  $F_{N-k, k-1}$
- C.  ~~$F_{k-1, N-k}$~~
- D.  $F_{k, N-1}$

**Correct answer: C****EXAM SHORTCUT**

Numerator df = treatment df =  $k-1$ ; denominator df = error df =  $N-k$ .

**Q6.**

Compared to Tukey's HSD, Scheffe's method for multiple comparisons is

- A. valid only when comparing exactly two treatments
- B. always identical to Tukey's HSD
- C. less conservative than Tukey's HSD for pairwise comparisons
- D. ~~more~~ **conservative for simple pairwise comparisons, but valid for testing ANY linear contrast, not just pairwise ones**

**Correct answer: D****EXAM SHORTCUT**

Scheffe controls the family-wise error rate over the INFINITE family of all contrasts, so it must be wider (more conservative) than Tukey for the narrower pairwise-only case.

# Estimable Functions and Estimability Conditions

18 forecast questions in this topic

Linear Models

Estimable Functions and Estimability Conditions

Medium

**Q1.**

Consider the Gauss–Markov model  $E(Y_1) = \beta_1 + \beta_2 + \beta_3$ ,  $E(Y_2) = \beta_1 - \beta_3$ ,  $E(Y_3) = 2\beta_1 + \beta_2$ , with uncorrelated  $Y_i$  having common variance  $\sigma^2$ . The function  $\lambda_1\beta_1 + \lambda_2\beta_2 + \lambda_3\beta_3$  is estimable if and only if

**A.  $\lambda_1 + \lambda_3 = 2\lambda_2$**

B.  $\lambda_1 + \lambda_2 = 2\lambda_3$

C.  $\lambda_1 + \lambda_2 + \lambda_3 = 0$

D.  $\lambda_2 + \lambda_3 = 2\lambda_1$

**Correct answer: A**

## EXAM SHORTCUT

Row 3 = row 1 + row 2, so the row space is spanned by (1, 1, 1) and (1, 0, -1);  $\lambda = (a + b, a, a - b)$  gives  $\lambda_1 + \lambda_3 = 2\lambda_2$ .

Linear Models

Estimable Functions and Estimability Conditions

Medium

**Q2.**

Let  $Y_1, Y_2, Y_3$  be uncorrelated with common variance and  $E(Y_1) = \theta_1 + \theta_2$ ,  $E(Y_2) = \theta_2 + \theta_3$ ,  $E(Y_3) = \theta_1 + 2\theta_2 + \theta_3$ . For estimability of  $l_1\theta_1 + l_2\theta_2 + l_3\theta_3$ , consider the following conditions: 1.  $l_1 = l_2 + l_3$  2.  $l_2 = l_1 + l_3$  3.  $l_3 = l_1 + l_2$  Which of the above condition(s) is/are correct?

A. 1, 2 and 3

B. 3 only

C. 1 only

**D. 2 only**

**Correct answer: D**

## EXAM SHORTCUT

Row 3 = row 1 + row 2; row space spanned by (1, 1, 0), (0, 1, 1), so  $\lambda = (a, a + b, b)$ , i.e.,  $l_2 = l_1 + l_3$ .

Linear Models

Estimable Functions and Estimability Conditions

Medium

**Q3.**

For the one-way classification model  $y_{ij} = \mu + \tau_i + \epsilon_{ij}$ ,  $i = 1, 2, 3$ ;  $j = 1, 2, \dots, n_i$ , which one of the following parametric functions is estimable?

A.  $\mu$

B.  $\tau_1 + \tau_2 + \tau_3$

C.  $\tau_1$

**D.  $\tau_1 - \tau_3$**

**Correct answer: D**

## EXAM SHORTCUT

Estimable iff coefficient of  $\mu$  equals sum of  $\tau$ -coefficients; only  $\tau_1 - \tau_3$  (a contrast) satisfies  $0 = 1 - 1$ .

Q4.

In the model  $y_{ij} = \mu + \tau_i + \epsilon_{ij}$ ,  $i = 1, 2, 3, 4$ ;  $j = 1, 2, \dots, n_i$ , which one of the following is not estimable?

A.  $\tau_1 + \tau_2$ B.  $\mu + \tau_1$ C.  $\tau_1 - \tau_2$ D.  $\mu + (1/4)(\tau_1 + \tau_2 + \tau_3 + \tau_4)$ 

Correct answer: A

## EXAM SHORTCUT

$\tau_1 + \tau_2$  needs  $\mu$ -coefficient = 2; it is 0. Items (b), (c), (d) all satisfy  $\lambda_0 = \sum i\lambda_i$ .

Q5.

Consider the two-way additive model  $y_{ij} = \mu + \alpha_i + \beta_j + \epsilon_{ij}$ ,  $i = 1, 2$ ;  $j = 1, 2$ . For which one of the following choices of  $\lambda'$  is the function  $\lambda'\theta$ , where  $\theta' = (\mu, \alpha_1, \alpha_2, \beta_1, \beta_2)$ , not estimable?

A. (0, 1, -1, 0, 0)

B. (1, 1, 0, 0, 1)

C.  $(0, 0, 0, 1, 1)$ 

D. (1, 0, 1, 1, 0)

Correct answer: C

## EXAM SHORTCUT

Rule:  $\lambda_0 = \sum i\alpha_i = \sum j\beta_j$ . For (0, 0, 0, 1, 1):  $\sum j\beta_j = 2 \neq 0$ , so  $\beta_1 + \beta_2$  is not estimable.

Q6.

Consider the model  $E(Y_{ij}) = \alpha_i + \beta_j$ ;  $i = 1, 2, 3$ ;  $j = 1, 2$ , where the  $Y_{ij}$  are uncorrelated with common variance. The function  $l_1\alpha_1 + l_2\alpha_2 + l_3\alpha_3 + m_1\beta_1 + m_2\beta_2$  is estimable if and only if

A.  $l_1 + l_2 + l_3 = m_1 + m_2$ B.  $l_1 = l_2 = l_3$ C.  $l_1 + l_2 + l_3 + m_1 + m_2 = 0$ D.  $l_1 + l_2 + l_3 + m_1 + m_2 = 1$ 

Correct answer: A

## EXAM SHORTCUT

Each row of  $X$  has  $\alpha$ -coefficients summing to 1 and  $\beta$ -coefficients summing to 1; the row space is exactly  $\{(l, m) : \sum l_i = \sum m_j\}$ .

Q7.

In the linear model  $Y = X\beta + \epsilon$ , consider the following statements: 1. If  $X$  has full column rank, then every linear parametric function is estimable. 2. A function  $\lambda'\beta$  is estimable if and only if  $\lambda' = c'X$  for some vector  $c$ . Which of the above statements is/are correct?

- A. Neither 1 nor 2
- B. 1 only
- C. Both 1 and 2
- D. 2 only

Correct answer: C

## EXAM SHORTCUT

Both are standard characterizations: full column rank  $\Rightarrow$  all LPFs estimable;  $\lambda'\beta$  estimable  $\iff \lambda \in \mathcal{C}(X')$ .

Q8.

In a Gauss–Markov model with  $n = 10$  observations,  $p = 6$  parameters and  $\text{rank}(X) = 4$ , the maximum number of linearly independent estimable parametric functions is

- A. 10
- B. 2
- C. 4
- D. 6

Correct answer: C

## EXAM SHORTCUT

The number of linearly independent estimable functions equals  $\text{rank}(X) = 4$ .

Q9.

For the model  $Y = X\beta + \epsilon$  with  $X$  of order  $n \times p$  and rank  $r$ , consider the following statements: I. The set of all vectors  $\lambda$  for which  $\lambda'\beta$  is estimable is a vector subspace of  $\mathbb{R}^p$ . II. The dimension of this subspace is  $n - r$ . Which of the statements given above is/are correct?

- A. Neither I nor II
- B. I only
- C. Both I and II
- D. II only

Correct answer: B

## EXAM SHORTCUT

The estimable  $\lambda$ 's form the row space of  $X$ , a subspace of dimension  $r$  (not  $n - r$ ).

## Q10.

Let  $E(Y_1) = \beta_1 - \beta_2$ ,  $E(Y_2) = \beta_2 - \beta_3$ ,  $E(Y_3) = \beta_3 - \beta_1$ , the  $Y_i$  being uncorrelated with common variance. Then  $\lambda_1\beta_1 + \lambda_2\beta_2 + \lambda_3\beta_3$  is estimable if and only if

- A.  $\lambda_1 + \lambda_2 + \lambda_3 = 1$
- B.  $\lambda_1 - \lambda_2 + \lambda_3 = 0$
- C.  $\lambda_1 + \lambda_2 + \lambda_3 = 0$
- D.  $\lambda_1 = \lambda_2 = \lambda_3$

Correct answer: C

## EXAM SHORTCUT

Rows sum to zero: row space =  $\{\lambda: \sum \lambda_i = 0\}$  (rank 2), i.e., exactly the contrasts.

## Q11.

For the model  $y_{ij} = \mu + \tau_i + \epsilon_{ij}$ ,  $i = 1, 2$ ;  $j = 1, 2, 3$ , consider the following statements: 1.  $\mu + \tau_2$  is estimable. 2.  $\tau_1 - \tau_2$  is estimable. 3.  $\mu$  is estimable. Which of the above statements are correct?

- A. 2 and 3 only
- B. 1 and 3 only
- C. 1, 2 and 3
- D. 1 and 2 only

Correct answer: D

## EXAM SHORTCUT

$\mu + \tau_2 = E(y_{2j})$  and  $\tau_1 - \tau_2$  are estimable;  $\mu$  alone is not ( $1 \neq 1 + 0$  rule fails:  $\lambda_0 = 1$ ,  $\sum \lambda_i = 0$ ).

## Q12.

Consider a two-way classification with one observation per cell:  $y_{ij} = \mu + \alpha_i + \beta_j + \epsilon_{ij}$ ;  $i = 1, \dots, p$ ;  $j = 1, \dots, q$ . Which one of the following is estimable?

- A.  $\alpha_i$
- B.  $\mu$
- C.  $\alpha_1 + \beta_1$
- D.  $\beta_j - \beta_{j'}$ ,  $j \neq j'$

Correct answer: D

## EXAM SHORTCUT

Only contrasts among the  $\alpha$ 's or among the  $\beta$ 's (and cell expectations) are estimable;  $\mu$ ,  $\alpha_i$ ,  $\alpha_1 + \beta_1$  all fail the row-space condition.

## Q13.

Let  $E(Y_1) = \beta_1 + 2\beta_2$ ,  $E(Y_2) = 2\beta_1 + 4\beta_2$ ,  $E(Y_3) = 3\beta_1 + 6\beta_2$ , the  $Y_i$  being uncorrelated with common variance. Which one of the following is estimable?

- A.  $\beta_1$
- B.  $\beta_2$
- C.  $\beta_1 + 2\beta_2$
- D.  $2\beta_1 + \beta_2$

Correct answer: C

## EXAM SHORTCUT

$\text{rank}(X) = 1$ ; row space = multiples of  $(1, 2)$ , so only multiples of  $\beta_1 + 2\beta_2$  are estimable.

## Q14.

Let  $\lambda'\beta$  be estimable in the model  $Y = X\beta + \epsilon$ , where  $X$  is  $n \times p$  of rank  $k < p$ . Let  $\hat{\beta}$  be any solution of the normal equations  $X'X\hat{\beta} = X'Y$  and let  $r$  be any solution of  $X'Xr = \lambda$ . Consider the following statements: I.  $\lambda'\hat{\beta} = r'X'Y$  for every such  $\hat{\beta}$  and  $r$ . II. The common value is invariant to the choice of  $\hat{\beta}$  and  $r$ . Which of the statements given above is/are correct?

- A. Neither I nor II
- B. I only
- C. II only
- D. Both I and II

Correct answer: D

## EXAM SHORTCUT

$\lambda'\hat{\beta} = (X'Xr)'\hat{\beta} = r'X'X\hat{\beta} = r'X'Y$ ; estimability makes the value invariant to both choices.

## Q15.

Consider the following statements: I. If  $\lambda'\beta$  and  $\mu'\beta$  are estimable, then  $a\lambda'\beta + b\mu'\beta$  is estimable for all real  $a, b$ . II. If  $\lambda'\beta$  is not estimable, then no linear function of the observations is unbiased for it. Which of the statements given above is/are correct?

- A. I only
- B. II only
- C. Both I and II
- D. Neither I nor II

Correct answer: C

## EXAM SHORTCUT

Estimable functions form a linear space (I); non-estimability means no linear unbiased estimator exists, by definition (II).

**Q16.**

In the Gauss–Markov model  $Y = X\beta + \epsilon$  with  $X$  of order  $n \times p$  and  $\text{rank}(X) = r < p$ , consider the following statements: I. Each individual component  $\beta_j$  of  $\beta$  is estimable. II. Each  $E(Y_i)$ ,  $i = 1, \dots, n$ , is an estimable parametric function. Which of the statements given above is/are correct?

- A. Both I and II
- B. ~~A~~ I only
- C. Neither I nor II
- D. I only

**Correct answer: B**

**EXAM SHORTCUT**

With  $r < p$  not all  $\beta_j$  are estimable; but  $E(Y_i) = x_i'\beta$  is always estimable ( $Y_i$  itself is unbiased for it).

**Q17.**

Let  $E(Y_1) = 2\beta_1 + \beta_2$ ,  $E(Y_2) = \beta_1 + \beta_2$ ,  $E(Y_3) = 3\beta_1 + 2\beta_2$ , with the usual uncorrelated, equal variance assumptions. Then

- A. only multiples of  $\beta_1 + \beta_2$  are estimable
- B. only multiples of  $\beta_1 - \beta_2$  are estimable
- C. ~~A~~ every linear function  $\lambda_1\beta_1 + \lambda_2\beta_2$  is estimable
- D. no linear function is estimable

**Correct answer: C**

**EXAM SHORTCUT**

$\text{rank}(X) = 2 = p$  (rows 1 and 2 are independent), i.e., full column rank, so all linear functions are estimable — estimability depends on column rank, not on  $n$ .

**Q18.**

Consider the model with four uncorrelated observations  $E(Y_1) = \beta_1 + \beta_2 + 2\beta_3$ ,  $E(Y_2) = \beta_1 + \beta_3$ ,  $E(Y_3) = 2\beta_1 + \beta_2 + 3\beta_3$ ,  $E(Y_4) = \beta_2 + \beta_3$ . Then  $\lambda_1\beta_1 + \lambda_2\beta_2 + \lambda_3\beta_3$  is estimable if and only if

- A.  $\lambda_2 = \lambda_1 + \lambda_3$
- B.  $\lambda_1 + \lambda_2 + \lambda_3 = 0$
- C. ~~A~~  $\lambda_3 = \lambda_1 + \lambda_2$
- D.  $\lambda_1 = \lambda_2 + \lambda_3$

**Correct answer: C**

**EXAM SHORTCUT**

Rows  $3 = 1 + 2$  and  $4 = 1 - 2$ ; row space spanned by  $(1, 1, 2)$ ,  $(1, 0, 1)$ :  $\lambda = (a + b, a, 2a + b)$ , so  $\lambda_3 = \lambda_1 + \lambda_2$ .

# BLUES and Least-Squares Computation in Small Models

16 forecast questions in this topic

Linear Models

BLUES and Least-Squares Computation in Small Models

Medium

Q1.

Let  $Y_1, Y_2, Y_3$  be uncorrelated with common variance  $\sigma^2$  and  $E(Y_1) = \beta_1, E(Y_2) = \beta_2, E(Y_3) = \beta_1 + \beta_2$ . The BLUE of  $\beta_1$  is

- A.  $(Y_1 + Y_3)/(2)$
- B.  $(2Y_1 + Y_2 - Y_3)/(3)$
- C.  $\checkmark (2Y_1 - Y_2 + Y_3)/(3)$
- D.  $Y_1$

Correct answer: C

## EXAM SHORTCUT

$X'X = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ ,  $X'y = (y_1 + y_3, y_2 + y_3)'$ ;  $\hat{\beta}_1 = (1/3)[2(y_1 + y_3) - (y_2 + y_3)] = (2y_1 - y_2 + y_3)/3$ .

Linear Models

BLUES and Least-Squares Computation in Small Models

Medium

Q2.

In the model of F2.1, the BLUE of  $\beta_1 - \beta_2$  is

- A.  $(2(Y_1 - Y_2))/(3)$
- B.  $(Y_1 - Y_2 + Y_3)/(3)$
- C.  $Y_3 - Y_2$
- D.  $\checkmark Y_1 - Y_2$

Correct answer: D

## EXAM SHORTCUT

$\hat{\beta}_1 - \hat{\beta}_2 = (1)/(3)(3y_1 - 3y_2) = y_1 - y_2$ ; elegant cancellation of the  $y_3$  terms.

Linear Models

BLUES and Least-Squares Computation in Small Models

Medium

Q3.

Let  $E(Y_1) = \theta, E(Y_2) = 2\theta, E(Y_3) = \theta$ , with uncorrelated observations of common variance  $\sigma^2$ . The least-squares estimator of  $\theta$  and its variance are

- A.  $\checkmark (Y_1 + 2Y_2 + Y_3)/(6), (\sigma^2)/(6)$
- B.  $(Y_1 + Y_2 + Y_3)/(3), (\sigma^2)/(3)$
- C.  $(Y_1 + 2Y_2 + Y_3)/(4), (\sigma^2)/(6)$
- D.  $(Y_1 + Y_2 + Y_3)/(4), (\sigma^2)/(4)$

Correct answer: A

## EXAM SHORTCUT

$\hat{\theta} = \sum x_i y_i / \sum x_i^2$  with  $x = (1, 2, 1)$ :  $(y_1 + 2y_2 + y_3)/6$ ;  $V = \sigma^2/6$ .

Q4.

Consider  $y_i = \beta_0 + \beta_1 x_i + e_i$ ,  $i = 1, 2, 3$ , with  $x_1 = -1$ ,  $x_2 = 0$ ,  $x_3 = 1$  and the usual assumptions. The best linear unbiased estimates of  $(\beta_0, \beta_1)$  are

A.  $((y_2 - y_1)/2, y_3 - y_1)$ B.  $(y, (y_3 - y_1)/2)$ C.  $((y_1 + y_3)/2, y)$ D.  $(y, (y_2 - y_1)/2)$ 

Correct answer: B

## EXAM SHORTCUT

$x = 0$ , so  $\hat{\beta}_0 = y$  and  $\hat{\beta}_1 = S_{xy}/S_{xx} = (y_3 - y_1)/2$ .

Q5.

In the setting of F2.4, consider the following statements: I.  $\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = 0$ . II.  $V(\hat{\beta}_1) = (\sigma^2)/(2)$ . Which of the statements given above is/are correct?

A. Neither I nor II

B. Both I and II

C. I only

D. II only

Correct answer: B

## EXAM SHORTCUT

$x = 0 \Rightarrow \text{Cov} = 0$ ;  $V(\hat{\beta}_1) = \sigma^2/S_{xx} = \sigma^2/2$ .

Q6.

Let  $X_i, Y_i, Z_i$ ,  $i = 1, 2, 3$ , be nine independent observations with common variance  $\sigma^2$  and  $E(X_i) = \theta_1 + \theta_2$ ,  $E(Y_i) = \theta_1$ ,  $E(Z_i) = \theta_2 - \theta_1$ . Write  $X = \sum X_i$ ,  $Y = \sum Y_i$ ,  $Z = \sum Z_i$  (sums over  $i = 1$  to 3). The BLUE of  $\theta_1$  is

A.  $(1/9)[X + Y - Z]$ B.  $(1/9)[X - Y + Z]$ C.  $(1/6)[X + Z]$ D.  $(1/6)[X + Y - Z]$ 

Correct answer: A

## EXAM SHORTCUT

$X'X = \text{diag}(9, 6)$  (columns orthogonal),  $X'y = (X + Y - Z, X + Z)'$ , so  $\hat{\theta}_1 = (X + Y - Z)/9$ .

Q7.

In the setting of F2.6, the BLUE of  $\theta_2$  and its variance are

- A.  $(1/9)[X+Y+Z], \sigma^2/9$
- B.  $\checkmark(1/6)[X+Z], \sigma^2/6$
- C.  $(1/9)[X+Z], \sigma^2/9$
- D.  $(1/6)[X-Z], \sigma^2/6$

Correct answer: B

## EXAM SHORTCUT

$\hat{\theta}_2 = (X+Z)/6$ ,  $V(\hat{\theta}_2) = \sigma^2/6$  from  $(X'X)^{-1} = \text{diag}(1/9, 1/6)$ .

Q8.

Consider the model  $Y_1 = A+B+C+D+e_1$ ,  $Y_2 = A-B+C-D+e_2$ ,  $Y_3 = A+B-C-D+e_3$ ,  $Y_4 = A-B-C+D+e_4$ . If this model is written as  $Y = X\beta + e$ , then  $(X'X)^{-1}$  equals

- A.  $(1/2)I_4$
- B.  $\checkmark(1/4)I_4$
- C.  $4I_4$
- D.  $(1/3)I_4$

Correct answer: B

## EXAM SHORTCUT

The four  $\pm 1$  columns are mutually orthogonal, so  $X'X = 4I_4$  and  $(X'X)^{-1} = I_4/4$ .

Q9.

In the model of F2.8, the least-squares estimator of  $B$  is

- A.  $(Y_1 + Y_2 + Y_3 + Y_4)/(4)$
- B.  $\checkmark(Y_1 - Y_2 + Y_3 - Y_4)/(4)$
- C.  $(Y_1 - Y_2 - Y_3 + Y_4)/(4)$
- D.  $(Y_1 + Y_2 - Y_3 - Y_4)/(4)$

Correct answer: B

## EXAM SHORTCUT

Column of  $B$  is  $(1, -1, 1, -1)'$ ;  $\hat{B} = (1)/(4)(Y_1 - Y_2 + Y_3 - Y_4)$ .

## Q10.

Let  $Y_i = i\beta + u_i$ ,  $i = 1, 2, 3, 4$ , where the  $u_i$  are independent with  $E(u_i) = 0$  and  $V(u_i) = i\sigma^2$ . The best linear unbiased estimator of  $\beta$  is

- A.  $(1)/(10)(Y_1 + Y_2 + Y_3 + Y_4)$   
 B.  $(1)/(4)(Y_1 + Y_2 + Y_3 + Y_4)$   
 C.  $(1)/(10)(Y_1 + 2Y_2 + 3Y_3 + 4Y_4)$   
 D.  $(1)/(30)(Y_1 + 2Y_2 + 3Y_3 + 4Y_4)$

Correct answer: A

## EXAM SHORTCUT

Weighted LS:  $\beta = \Sigma(x_i y_i / v_i) / \Sigma(x_i^2 / v_i) = \Sigma y_i / \Sigma i = (Y_1 + \dots + Y_4) / 10$ .

## Q11.

Let  $Y_i = i\beta + u_i$ ,  $i = 1, 2, 3$ , with independent  $u_i$ ,  $E(u_i) = 0$ ,  $V(u_i) = i^2\sigma^2$ . The BLUE of  $\beta$  and its variance are

- A.  $(6Y_1 + 3Y_2 + 2Y_3)/(18), (\sigma^2)/(3)$   
 B.  $(Y_1 + Y_2 + Y_3)/(6), (\sigma^2)/(6)$   
 C.  $(6Y_1 + 3Y_2 + 2Y_3)/(11), (\sigma^2)/(11)$   
 D.  $(Y_1 + 2Y_2 + 3Y_3)/(14), (\sigma^2)/(14)$

Correct answer: A

## EXAM SHORTCUT

Weights  $1/i^2$ :  $\beta = (1/3)\Sigma(Y_i/i) = (6Y_1 + 3Y_2 + 2Y_3)/18$ ;  $V = \sigma^2/\Sigma(x_i^2/v_i) = \sigma^2/3$ .

## Q12.

Two treatments are applied to  $n_1 = 4$  and  $n_2 = 6$  experimental units respectively, all observations being uncorrelated with common variance  $\sigma^2$ , and  $E(y_{1j}) = \mu_1$ ,  $E(y_{2j}) = \mu_2$ . The variance of the BLUE of  $\mu_1 - \mu_2$  is

- A.  $(5\sigma^2)/(12)$   
 B.  $(\sigma^2)/(10)$   
 C.  $(2\sigma^2)/(5)$   
 D.  $(\sigma^2)/(24)$

Correct answer: A

## EXAM SHORTCUT

$V(y_1 - y_2) = \sigma^2(1/4 + 1/6) = 5\sigma^2/12$ .

## Q13.

Let  $E(Y_1) = \beta_1 + \beta_2$ ,  $E(Y_2) = \beta_1 - \beta_2$ ,  $E(Y_3) = \beta_1 + 2\beta_2$ , with uncorrelated observations of common variance  $\sigma^2$ . The BLUE of  $\beta_1$  is

- A.  $(2Y_1 + 4Y_2 + Y_3)/7$   
 B.  $(Y_1 + Y_2)/2$   
 C.  $(4Y_1 + 2Y_2 + Y_3)/7$   
 D.  $(Y_1 + Y_2 + Y_3)/3$

Correct answer: A

## EXAM SHORTCUT

$X'X = \begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix}$ ;  $\beta_1 = (1/14)[6\sum y_i - 2(y_1 - y_2 + 2y_3)] = (2y_1 + 4y_2 + y_3)/7$ ; check unbiasedness:  $E = (2+4+1)\beta_1/7 = \beta_1$ .

## Q14.

Let  $E(Y_1) = \beta_1$ ,  $E(Y_2) = \beta_1 + \beta_2$ ,  $E(Y_3) = \beta_1 + \beta_2 + \beta_3$ , the  $Y_i$  being uncorrelated with common variance  $\sigma^2$ . Consider the following statements: I. The residual sum of squares of the least-squares fit is identically zero. II.  $\sigma^2$  cannot be unbiasedly estimated from the least-squares residuals. Which of the statements given above is/are correct?

- A. I only  
 B. II only  
 C. Neither I nor II  
 D. Both I and II

Correct answer: D

## EXAM SHORTCUT

$n = p = 3$  with full rank: the fit is exact,  $RSS = 0$  with  $n - r = 0$  degrees of freedom, so no unbiased residual-based estimator of  $\sigma^2$  exists.

## Q15.

Consider  $Y_i = \mu + (-1)^i \delta + u_i$ ,  $i = 1, 2, \dots, 2n$ , with independent errors,  $E(u_i) = 0$ ,  $V(u_i) = \sigma^2$ . The least-squares estimators are: I.  $\hat{\mu} = (1)/(2n) \sum_{i=1}^{2n} Y_i$  II.  $\hat{\delta} = (1)/(2n) [\sum_{i=1}^n Y_{2i} - \sum_{i=1}^n Y_{2i-1}]$  Which of the statements given above is/are correct?

- A. Both I and II  
 B. Neither I nor II  
 C. II only  
 D. I only

Correct answer: A

## EXAM SHORTCUT

The two columns (all ones; alternating  $\pm 1$ ) are orthogonal with squared lengths  $2n$ , giving exactly the stated estimators.

**Q16.**

Let  $\lambda'\beta$  be estimable and  $\hat{\beta}$  any solution of the normal equations. Consider the following statements: I.  $\lambda'\hat{\beta}$  is the best linear unbiased estimator of  $\lambda'\beta$ . II.  $\lambda'\hat{\beta}$  takes the same value for every solution  $\hat{\beta}$  of the normal equations.

Which of the statements given above is/are correct?

- A. I only
- B. Neither I nor II
- C. II only
- D. Both I and II

**Correct answer: D**

**EXAM SHORTCUT**

Gauss–Markov theorem plus invariance of  $\lambda'\hat{\beta}$  for estimable  $\lambda'\beta$ .

# Orthogonality: When Are BLUEs Uncorrelated?

10 forecast questions in this topic

Linear Models

Orthogonality: When Are BLUEs Uncorrelated?

Medium

Q1.

Let  $E(y_1) = 2\beta_1 + \beta_2$ ,  $E(y_2) = \beta_1 - \beta_2$ ,  $E(y_3) = \beta_1 + \alpha\beta_2$ , with uncorrelated observations of constant variance. The value of  $\alpha$  for which the BLUEs of  $\beta_1$  and  $\beta_2$  are uncorrelated is

A.  $-1$

B. 0

C. +2

D. +1

Correct answer: A

## EXAM SHORTCUT

Columns  $(2, 1, 1)$  and  $(1, -1, \alpha)$ :  $2 - 1 + \alpha = 0 \Rightarrow \alpha = -1$ .

Linear Models

Orthogonality: When Are BLUEs Uncorrelated?

Medium

Q2.

Let  $E(y_1) = \beta_1 + 2\beta_2$ ,  $E(y_2) = 2\beta_1 + \beta_2$ ,  $E(y_3) = \beta_1 + \alpha\beta_2$ , with the usual assumptions. The least-squares estimators of  $\beta_1$  and  $\beta_2$  are uncorrelated when  $\alpha$  equals

A.  $-2$

B. 0

C. +4

D.  $-4$

Correct answer: D

## EXAM SHORTCUT

$(1, 2, 1) \cdot (2, 1, \alpha) = 2 + 2 + \alpha = 0 \Rightarrow \alpha = -4$ .

Linear Models

Orthogonality: When Are BLUEs Uncorrelated?

Medium

Q3.

Let  $E(y_1) = \beta_1 + \beta_2$ ,  $E(y_2) = 2\beta_1 + \alpha\beta_2$ ,  $E(y_3) = \beta_1 - \beta_2$ , with uncorrelated equal-variance observations. The BLUEs of  $\beta_1$  and  $\beta_2$  are uncorrelated when  $\alpha$  equals

A.  $-1$

B. +2

C. +1

D. 0

Correct answer: D

## EXAM SHORTCUT

$(1, 2, 1) \cdot (1, \alpha, -1) = 1 + 2\alpha - 1 = 2\alpha = 0 \Rightarrow \alpha = 0$ .

Q4.

Let  $E(y_i) = \beta_1 + c_i\beta_2$ ,  $i = 1, 2, 3, 4$ , with  $c_1 = 1$ ,  $c_2 = -1$ ,  $c_3 = 1$ ,  $c_4 = \alpha$ , under the usual assumptions. The value of  $\alpha$  making  $\hat{\beta}_1$  and  $\hat{\beta}_2$  uncorrelated is

- A. 0
- B. -3
- C. +1
- D. -1

Correct answer: D

## EXAM SHORTCUT

$(1, 1, 1, 1) \cdot (1, -1, 1, \alpha) = 1 + \alpha = 0 \Rightarrow \alpha = -1$ .

Q5.

For a full-rank model with two parameters, consider the following statements: I.  $\text{Cov}(\hat{\beta}_1, \hat{\beta}_2) = 0$  if and only if the two columns of  $X$  are orthogonal. II. If  $X'X$  is diagonal, then  $(X'X)^{-1}$  is diagonal. Which of the statements given above is/are correct?

- A. Neither I nor II
- B. I only
- C. Both I and II
- D. II only

Correct answer: C

## EXAM SHORTCUT

For  $2 \times 2$  matrices the off-diagonal of the inverse vanishes exactly when the offdiagonal of  $X'X$  (the column inner product) vanishes; II is immediate.

Q6.

In the simple linear regression model  $y_i = \beta_0 + \beta_1 x_i + e_i$  with the usual assumptions,  $\hat{\beta}_0$  and  $\hat{\beta}_1$  are uncorrelated if and only if

- A.  $\sum x_i = 0$
- B.  $\sum y_i = 0$
- C.  $n$  is even
- D.  $\sum x_i^2 = 0$

Correct answer: A

## EXAM SHORTCUT

$\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\frac{\sum x_i^2}{S_{xx}}$ , zero iff  $\sum x_i = 0$ .

Q7.

Independent observations follow  $y_i = \beta_0 + \beta_1 x_i + u_i$ ,  $i = 1, \dots, 10$ , with error variance  $\sigma^2 = 2$ . Given  $\sum x_i = 50$  and  $\sum x_i^2 = 290$ , the covariance between the least-squares estimators of  $\beta_0$  and  $\beta_1$  is

- A. 0
- B. 0.25
- C. ~~-0.25~~
- D. -0.125

Correct answer: C

## EXAM SHORTCUT

$\bar{x} = 5$ ,  $S_{xx} = 290 - 10(25) = 40$ ;  $\text{Cov} = -\bar{x}\sigma^2/S_{xx} = -5(2)/40 = -0.25$ .

Q8.

If the inner product of every pair of distinct columns of the design matrix  $X$  equals zero, the design is called

- A. a non-orthogonal design
- B. an incomplete block design
- C. ~~An orthogonal design~~
- D. a saturated design

Correct answer: C

## EXAM SHORTCUT

Definition of an orthogonal design (2025 Q53 archetype).

Q9.

For an orthogonal design (as in F3.8) with columns  $c_1, \dots, c_p$ , consider the following statements: I.  $\hat{\beta}_j = c_j'/(j'y)/(c_j'jc_j)$ , and it is unchanged if any other factor is dropped from the model. II.  $V(\hat{\beta}_j) = \sigma^2/(c_j'jc_j)$ . Which of the statements given above is/are correct?

- A. I only
- B. Neither I nor II
- C. II only
- D. ~~Both I and II~~

Correct answer: D

## EXAM SHORTCUT

Diagonal  $X'X$  decouples the normal equations; both statements are the standard consequences.

**Q10.**

Let  $E(y_1) = \beta_1 + \beta_2$ ,  $E(y_2) = \beta_1 - \beta_2$ ,  $E(y_3) = \alpha\beta_1$ , with uncorrelated equal-variance observations. The BLUEs of  $\beta_1$  and  $\beta_2$  are uncorrelated

- A. only when  $\alpha = 1$
- B. only when  $\alpha = 0$
- C. only when  $\alpha = -1$
- D. **for every value of  $\alpha$**

**Correct answer: D**

**EXAM SHORTCUT**

$(1, 1, \alpha) \cdot (1, -1, 0) = 0$  identically — the columns are orthogonal for every  $\alpha$ .

# Estimation of Error Variance

12 forecast questions in this topic

Linear Models

Estimation of Error Variance

Medium

**Q1.**

Let  $Y_1, Y_2, Y_3, Y_4$  be uncorrelated with common variance  $\sigma^2$ ,  $E(Y_1) = E(Y_2) = \beta_1 + \beta_2 + \beta_3$ ,  $E(Y_3) = E(Y_4) = \beta_1 - \beta_2$ . Define  $e_1 = \sqrt{(1/2)}(Y_1 - Y_2)$  and  $e_2 = \sqrt{(1/2)}(Y_3 - Y_4)$ . An unbiased estimator of  $\sigma^2$  is

A.  $(1/2)(e_1^2 - e_2^2)$

B.  $(1/4)(e_1^2 + e_2^2)$

C.  $e_1^2 + e_2^2$

D.  $\sqrt{(1/2)}(e_1^2 + e_2^2)$

**Correct answer: D**

## EXAM SHORTCUT

$e_1, e_2$  have mean 0, variance  $\sigma^2$ , so  $E(e_1^2 + e_2^2) = 2\sigma^2$  (2019 Q4 / 2026 Q54 archetype).

Linear Models

Estimation of Error Variance

Medium

**Q2.**

Let  $Y_1, \dots, Y_6$  be uncorrelated with common variance  $\sigma^2$  and  $E(Y_{2i-1}) = E(Y_{2i}) = \mu_i$ ,  $i = 1, 2, 3$ . With  $e_i = \sqrt{(1/2)}(Y_{2i-1} - Y_{2i})$ , an unbiased estimator of  $\sigma^2$  is

A.  $e_1^2 + e_2^2 + e_3^2$

B.  $(e_1^2 + e_2^2 + e_3^2)/6$

C.  $\sqrt{(e_1^2 + e_2^2 + e_3^2)}/3$

D.  $(e_1^2 + e_2^2 + e_3^2)/2$

**Correct answer: C**

## EXAM SHORTCUT

Three error contrasts, each with  $E(e_i^2) = \sigma^2$ : divide the sum by 3.

Linear Models

Estimation of Error Variance

Medium

**Q3.**

In the model of F4.1, the number of degrees of freedom available for estimating  $\sigma^2$  is

A. 3

B. 1

C. 2

D. 4

**Correct answer: C**

## EXAM SHORTCUT

$n = 4$ , mean structure has rank 2 (rows  $(1, 1, 1)$  and  $(1, -1, 0)$ ), so  $n - r = 2$ .

## Q4.

For 4 observations from the model  $y_i = \beta_0 + \beta_1 x_i + e_i$  it is observed that  $\sum x_i = 0$ ,  $\sum y_i = 8$ ,  $\sum x_i y_i = 10$ ,  $\sum x_i^2 = 20$ ,  $\sum y_i^2 = 23$ . The estimated error variance is

- A. 0.667
- B. 2.0
- C. 1.0
- D. 0.5

Correct answer: C

## EXAM SHORTCUT

$\hat{\beta}_1 = 10/20 = 0.5$ ;  $S_{yy} = 23 - 4(22) = 7$ ;  $RSS = 7 - 0.5(10) = 2$ ;  $\hat{\sigma}^2 = 2/(4 - 2) = 1$ .

## Q5.

In a one-way classification with  $k = 3$  groups and  $N = 15$  observations in all, the within-group sum of squares is 48. The unbiased estimate of the error variance  $\sigma^2$  is

- A. 16
- B. 3.2
- C. 3.43
- D. 4

Correct answer: D

## EXAM SHORTCUT

$MSE = SSW/(N - k) = 48/12 = 4$ .

## Q6.

In a Gauss–Markov model with  $n = 12$  observations and  $\text{rank}(X) = 4$ , the expected value of the residual sum of squares is

- A.  $8\sigma^2$
- B.  $11\sigma^2$
- C.  $4\sigma^2$
- D.  $12\sigma^2$

Correct answer: A

## EXAM SHORTCUT

$E(RSS) = (n - r)\sigma^2 = 8\sigma^2$ .

Q7.

Under the normal Gauss–Markov model of full rank  $p$ , consider the following statements: I. The maximum likelihood estimator of  $\sigma^2$  is  $\text{RSS}/n$ , which is biased. II.  $\text{RSS}/(n-p)$  is unbiased for  $\sigma^2$ . Which of the statements given above is/are correct?

A. ~~Both I and II~~

B. Neither I nor II

C. II only

D. I only

Correct answer: A

## EXAM SHORTCUT

Classical results for the normal linear model.

Q8.

For a linear model fit,  $y'y = 30$  and  $\hat{\beta}'X'y = 22$ , with  $n-r = 4$ . The unbiased estimate of  $\sigma^2$  is

A. 8

B. ~~2~~

C. 7.5

D. 5.5

Correct answer: B

## EXAM SHORTCUT

$$\text{RSS} = y'y - \hat{\beta}'X'y = 8; \hat{\sigma}^2 = 8/4 = 2.$$

Q9.

Let  $E(y_1) = \beta_1$ ,  $E(y_2) = \beta_2$ ,  $E(y_3) = \beta_1 + \beta_2$  with uncorrelated equal-variance observations. The residual sum of squares of the least-squares fit is

A.  $(1/3)(y_1 + y_2 + y_3)^2$ B.  $(1/2)(y_1 + y_2 - y_3)^2$ C.  ~~$(1/3)(y_1 + y_2 - y_3)^2$~~ D.  $(1/3)(y_1 - y_2 + y_3)^2$ 

Correct answer: C

## EXAM SHORTCUT

All three residuals equal  $\pm(y_1 + y_2 - y_3)/3$ , giving  $\text{RSS} = (y_1 + y_2 - y_3)^2/3$  (2022 Q30 archetype).

## Q10.

In the full-rank model  $Y = X\beta + \epsilon$  with  $p$  parameters, which one of the following is an unbiased estimator of  $\sigma^2$ ?

- A.  $((Y - \hat{X}\beta)'(Y - \hat{X}\beta))/(n)$
- B.  $((Y - \hat{X}\beta)'(Y - \hat{X}\beta))/(p)$
- C.  $((Y - \hat{X}\beta)'(Y - \hat{X}\beta))/(n) - 1$
- D.  $((Y - \hat{X}\beta)'(Y - \hat{X}\beta))/(n - p)$

Correct answer: D

## EXAM SHORTCUT

RSS has  $n - p$  degrees of freedom in the full-rank case.

## Q11.

Under normality, with  $n - r = 8$  error degrees of freedom, the variance of the unbiased estimator  $\sigma^2 = \text{RSS}/8$  is

- A.  $(\sigma^4)/(8)$
- B.  $2\sigma^4$
- C.  $(\sigma^4)/(16)$
- D.  $(\sigma^4)/(4)$

Correct answer: D

## EXAM SHORTCUT

$\text{RSS}/\sigma^2 \sim \chi^2_8$ , so  $V(\hat{\sigma}^2) = 2\sigma^4/8 = \sigma^4/4$ .

## Q12.

In a one-way fixed-effects ANOVA, consider the following statements: I. The error mean square MSE is unbiased for  $\sigma^2$  whether or not  $H_0$  is true, and without assuming normality. II. The treatment mean square MST is unbiased for  $\sigma^2$  only when all treatment effects are equal. Which of the statements given above is/are correct?

- A. Neither I nor II
- B. II only
- C. Both I and II
- D. I only

Correct answer: C

## EXAM SHORTCUT

$E(\text{MSE}) = \sigma^2$  always;  $E(\text{MST}) = \sigma^2 + \text{positive term}$  unless all effects coincide.

# Generalized Inverse and Allied Matrix Theory

22 forecast questions in this topic

Linear Models

Generalized Inverse and Allied Matrix Theory

Medium

Q1.

A matrix  $G$  is called a generalized inverse of  $A$  if

- A.  $GA = I$
- B.  $AG = GA$
- C.  $GAG = G$
- D.  $AGA = A$

Correct answer: D

## EXAM SHORTCUT

Defining property; (c) alone defines an outer inverse, both together a reflexive inverse.

Linear Models

Generalized Inverse and Allied Matrix Theory

Medium

Q2.

Let  $A$  be a matrix of order  $5 \times 8$ . Then any generalized inverse of  $A$  is of order

- A.  $8 \times 5$
- B.  $8 \times 8$
- C.  $5 \times 5$
- D.  $5 \times 8$

Correct answer: A

## EXAM SHORTCUT

$A_{(m \times n)} \Rightarrow A^+_{(n \times m)}$ : conformability of  $AA^+$ .

Linear Models

Generalized Inverse and Allied Matrix Theory

Medium

Q3.

Consider the following statements: I. Every matrix (of any order, any rank) possesses at least one generalized inverse. II. The generalized inverse of  $A$  is unique if and only if  $A$  is square and non-singular, in which case  $A^+ = A^{-1}$ . Which of the statements given above is/are correct?

- A. Both I and II
- B. II only
- C. I only
- D. Neither I nor II

Correct answer: A

## EXAM SHORTCUT

Existence always holds; uniqueness exactly in the non-singular case.

**Q4.**

Consider the following statements in respect of a symmetric matrix  $A$ : I. A generalized inverse of  $A$  is not necessarily symmetric. II. A symmetric generalized inverse of  $A$  can always be found. Which of the statements given above is/are correct?

- A. II only
- B. Both I and II**
- C. I only
- D. Neither I nor II

**Correct answer: B****EXAM SHORTCUT**

The verbatim 2021 Q7 / 2023 Q63 / 2024 Q51 fact pair: both statements are true.

**Q5.**

Consider the following four conditions on a matrix  $G$  relative to  $A$ : 1.  $AGA = A$ ; 2.  $GAG = G$ ; 3.  $(AG)' = AG$ ; 4.  $(GA)' = GA$ . How many of these conditions must  $G$  satisfy in order to be the Moore–Penrose inverse of  $A$ ?

- A. Conditions 1 and 2 only
- B. Conditions 1, 2 and 3 only
- C. All four conditions**
- D. Only condition 1

**Correct answer: C****EXAM SHORTCUT**

Moore–Penrose requires all four Penrose conditions; any  $G$  with all four is unique.

**Q6.**

Consider the following statements: I. The Moore–Penrose inverse of any matrix exists and is unique. II. A matrix of less than full rank has infinitely many generalized inverses. Which of the statements given above is/are correct?

- A. I only
- B. II only
- C. Neither I nor II
- D. Both I and II**

**Correct answer: D****EXAM SHORTCUT**

MP inverse: existence and uniqueness are classical; rank-deficient matrices admit infinitely many  $g$ -inverses.

Q7.

If  $A^-$  is any generalized inverse of  $A$  and  $H = A^-A$ , consider the following statements: 1.  $H$  is idempotent. 2.  $AH = A$ . 3.  $\text{rank}(A) = \text{rank}(H) = \text{trace}(H)$ . Which of the above statements are correct?

A. 1 and 2 only

B. **1, 2 and 3**

C. 2 and 3 only

D. 1 and 3 only

**Correct answer: B****EXAM SHORTCUT**

$H^2 = A^-(AA^-A) = A^-A = H$ ;  $AH = AA^-A = A$ ; idempotency gives  $\text{rank} = \text{trace} = \text{rank}(A)$ .

Q8.

Let  $A$  be an  $n \times p$  matrix of rank  $r$  and  $A^-$  any generalized inverse of  $A$ . Consider the following statements: I.  $\text{trace}(AA^-) = \text{trace}(A^-A)$ . II.  $\text{trace}(AA^-) = r$ . Which of the statements given above is/are correct?

A. **Both I and II**

B. Neither I nor II

C. I only

D. II only

**Correct answer: A****EXAM SHORTCUT**

Both traces equal  $\text{rank}(A) = r$  since  $AA^-$  and  $A^-A$  are idempotent with rank  $r$ .

Q9.

Consider the following statements: I.  $\text{rank}(A^-) \geq \text{rank}(A)$  for every generalized inverse  $A^-$  of  $A$ . II.  $\text{rank}(A^-) = \text{rank}(A)$  if and only if  $A^-$  is a reflexive generalized inverse (i.e.,  $A^-AA^- = A^-$ ). Which of the statements given above is/are correct?

A. **Both I and II**

B. Neither I nor II

C. I only

D. II only

**Correct answer: A****EXAM SHORTCUT**

Standard rank inequality; equality characterizes reflexive inverses (2026 Q53 archetype).

**Q10.**

Consider the following statements: I. If  $A^-$  is a generalized inverse of  $A$ , then  $(A^-)'$  is a generalized inverse of  $A'$ . II. If  $A$  is symmetric and  $A^-$  is a generalized inverse of  $A$ , then  $(A^-)'$  is also a generalized inverse of  $A$ .

Which of the statements given above is/are correct?

**A. Both I and II**

B. Neither I nor II

C. II only

D. I only

**Correct answer: A**

**EXAM SHORTCUT**

$A'(A^-)'A' = (AA^-A)' = A'$ ; for symmetric  $A$  replace  $A'$  by  $A$ .

**Q11.**

Let  $A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 2 & 5 \end{bmatrix}$  and let  $A^-$  denote a generalized inverse of  $A$ . Consider the following statements: I.  $\text{rank}(A) = 2$ . II.  $\text{trace}(A^-A) = 2$ . III.  $\text{trace}(A) = 7$ . IV.  $A^-$  is unique. How many of the statements given above are correct?

A. Only one statement

B. Only two statements

C. All four statements

**D. Only three statements**

**Correct answer: D**

**EXAM SHORTCUT**

Row 3 = row 1 + row 2 so rank 2;  $\text{trace}(A^-A) = \text{rank} = 2$ ;  $\text{trace}(A) = 1+1+5 = 7$ ; a rank-deficient  $A$  has infinitely many  $A^-$  — I, II, III correct.

**Q12.**

Let  $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ . Which one of the following is not a generalized inverse of  $A$ ?

A.  $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

**B.  $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$**

C.  $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

D.  $\begin{bmatrix} 1/4 & 1/4 \\ 1/4 & 1/4 \end{bmatrix}$

**Correct answer: B**

**EXAM SHORTCUT**

For  $A = J_2$ ,  $AXA = (\sum_{ij} x_{ij} A_{ij})A$ , so  $X$  is a  $g$ -inverse iff its entries sum to 1; the identity's entries sum to 2.

## Q13.

For the matrices  $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ ,  $B = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$ , consider the following statements: I.  $ABA = A$ . II.  $BA$  is an idempotent matrix. III.  $\text{rank}(A) \neq \text{rank}(BA)$ . IV.  $\text{rank}(BA) = \text{trace}(BA)$ . How many of the statements given above are correct?

- A. Only one statement
- B. All four statements
- C. **Only three statements**
- D. Only two statements

**Correct answer: C**

## EXAM SHORTCUT

$B = A^{-1}$ , so  $ABA = A$ ,  $BA = I$  (idempotent), ranks equal (III false), and  $\text{rank}(BA) = \text{trace}(BA) = 2$ .

## Q14.

Let  $D = \text{diag}(3, 0, 2)$ . Which one of the following is not a generalized inverse of  $D$ ?

- A.  **$\text{diag}(1/3, 0, 2)$**
- B.  $\text{diag}(1/3, 7, 1/2)$
- C.  $\text{diag}(1/3, 0, 1/2)$
- D.  $\text{diag}(1/3, -4, 1/2)$

**Correct answer: A**

## EXAM SHORTCUT

Need  $d_{iix}d_{iix} = d_{ii}$ :  $x_{33}$  must be  $(1)/(2)$ , while the  $(2, 2)$  entry is free because  $d_{22} = 0$ .

## Q15.

The Moore–Penrose inverse of the column vector  $a = (1, 2, 2)'$  is

- A.  **$(1)/(9)(1, 2, 2)$**
- B.  $(1, 2, 2)$
- C.  $(1)/(5)(1, 2, 2)$
- D.  $(1)/(3)(1, 2, 2)'$

**Correct answer: A**

## EXAM SHORTCUT

$a^+ = a'/(a'a) = (1, 2, 2)/9$ ; check  $aa^+a = a$ .

**Q16.**

If the system of equations  $Ax = c$  has at least one solution, consider the following statements: I.  $\text{rank}(A) = \text{rank}(A, c)$ . II.  $AA^-c = c$  for every generalized inverse  $A^-$  of  $A$ . Which of the statements given above is/are correct?

- A. I only
- B. Neither I nor II
- C. II only
- D. Both I and II

**Correct answer: D****EXAM SHORTCUT**

Consistency  $\iff$  augmented rank condition  $\iff AA^-c = c$  (2024 Q53 archetype).

**Q17.**

If  $Ax = c$  is consistent and  $A^-$  is any generalized inverse of  $A$ , the general solution is

- A.  $x = (I - A^-A)c$
- B.  $x = A^-c$  only
- C.  $x = A^-c + A^-Az$ ,  $z$  arbitrary
- D.  $x = A^-c + (I - A^-A)z$ ,  $z$  arbitrary

**Correct answer: D****EXAM SHORTCUT**

Particular solution plus the general null-space element  $(I - A^-A)z$ .

**Q18.**

Let  $A$  be a matrix with 7 columns and rank 4, and let  $y$  be a non-null vector such that  $Ax = y$  is consistent. The number of linearly independent solutions of  $Ax = y$  is

- A. 4
- B. 7
- C. 3
- D. 5

**Correct answer: A****EXAM SHORTCUT**

For consistent  $Ax = y$ ,  $y \neq 0$ : maximum number of linearly independent solutions is  $n - r + 1 = 7 - 4 + 1 = 4$  (2026 Q51 archetype).

**Q19.**

In the linear model  $Y = X\beta + \epsilon$ , let  $G$  be any generalized inverse of  $X'X$ . Consider the following statements: I.  $XGX'$  is invariant to the choice of  $G$ . II.  $XGX'$  is symmetric only when  $G$  is symmetric. Which of the statements given above is/are correct?

- A. Both I and II
- B. I only
- C. II only
- D. Neither I nor II

**Correct answer: B**

**EXAM SHORTCUT**

$XGX'$  is the orthogonal projector on  $\mathcal{C}(X)$  for every  $G$ ; hence invariant and automatically symmetric — II is false.

**Q20.**

Let  $P = X(X'X)^{-1}X'$  with  $\text{rank}(X) = r$ . Consider the following statements: 1.  $P$  is idempotent. 2.  $PX = X$ . 3.  $\text{trace}(P) = r$ . Which of the above statements are correct?

- A. 1 and 3 only
- B. 1, 2 and 3
- C. 1 and 2 only
- D. 2 and 3 only

**Correct answer: B**

**EXAM SHORTCUT**

$P$  is the projector on  $\mathcal{C}(X)$ : idempotent, reproduces  $X$ ,  $\text{trace} = \text{rank} = r$ .

**Q21.**

Let  $M = I - A^-A$  for a generalized inverse  $A^-$  of  $A$ . Consider the following statements: 1.  $AM = 0$ . 2.  $M$  is idempotent. 3. Every solution of the homogeneous system  $Ax = 0$  can be written as  $Mz$  for some  $z$ . Which of the above statements are correct?

- A. 1, 2 and 3
- B. 2 and 3 only
- C. 1 and 2 only
- D. 1 and 3 only

**Correct answer: A**

**EXAM SHORTCUT**

$AM = A - AA^-A = 0$ ;  $M^2 = M$  using  $(A^-A)^2 = A^-A$ ; the null space of  $A$  is exactly the column space of  $M$ .

**Q22.**

Consider the following statements: I. If  $A$  is an idempotent matrix, then  $\text{rank}(A) = \text{trace}(A)$ . II. Given any matrix  $A$ , there exists a matrix  $A^-$  with  $AA^-A = A$  and  $\text{trace}(A^-A) = \text{rank}(A)$ . Which of the statements given above is/are correct?

**A.** I only**B. Both I and II****C.** Neither I nor II**D.** II only**Correct answer: B****EXAM SHORTCUT**

Idempotent  $\Rightarrow$  eigenvalues 0/1  $\Rightarrow$  rank = trace; and  $\text{trace}(A^-A) = \text{rank}(A)$  holds for every  $g$ -inverse (2025 Q55 archetype).

# Normal Equations and Their Solution

10 forecast questions in this topic

Linear Models

Normal Equations and Their Solution

Medium

Q1.

For the model  $Y = X\beta + e$  with  $E(e) = 0$ ,  $V(e) = \sigma^2 I$ , the least-squares normal equations are

A.  $\hat{X}\beta = Y$

B.  $X'X\hat{\beta} = X'Y$

C.  $X'\hat{Y}\beta = X'X$

D.  $XX'\beta = XY$

Correct answer: B

## EXAM SHORTCUT

Minimize  $(Y - X\beta)'(Y - X\beta)$ : differentiate to get  $X'X\hat{\beta} = X'Y$  (2022 Q2 archetype).

Linear Models

Normal Equations and Their Solution

Medium

Q2.

Consider the following statements: I. The normal equations  $X'X\hat{\beta} = X'Y$  are consistent for every observed  $Y$ , whatever the rank of  $X$ . II. The observational equations  $Y = X\beta$  are consistent for every observed  $Y$ . Which of the statements given above is/are correct?

A. I only

B. II only

C. Neither I nor II

D. Both I and II

Correct answer: A

## EXAM SHORTCUT

NE are always consistent since  $X'Y \in \mathcal{C}(X'X)$ ;  $Y = X\beta$  is consistent only when  $Y \in \mathcal{C}(X)$  (2025 Q57 key: II only there; here the statements are swapped).

Linear Models

Normal Equations and Their Solution

Medium

Q3.

If  $(X'X)^-$  is any generalized inverse of  $X'X$ , the general solution of the normal equations is  $\hat{\beta} =$

A.  $(X'X)^-X'Y + [I - (X'X)^-X'X]z$ , arbitrary

B.  $(X'X)^-X'Y$  only

C.  $X'Y + [I - (X'X)]z$ , arbitrary

D.  $(X'X)^-X'Y + (X'X)^-X'Xz$ , arbitrary

Correct answer: A

## EXAM SHORTCUT

Particular solution plus general solution of the homogeneous system.

Q4.

In the model  $Y = X\beta + \epsilon$  with  $X$  of order  $n \times p$  and  $\text{rank}(X) = r$ , consider the following statements: I. The normal equations have a unique solution if and only if  $r = p$ . II. If  $r < p$ , the normal equations have infinitely many solutions. Which of the statements given above is/are correct?

A. Both I and II

B. II only

C. I only

D. Neither I nor II

Correct answer: A

## EXAM SHORTCUT

$\text{rank}(X'X) = r$ : unique iff non-singular iff  $r = p$ ; otherwise an affine family of solutions.

Q5.

Let  $\hat{\beta}_1$  and  $\hat{\beta}_2$  be two different solutions of the normal equations. Consider the following statements: I.  $\hat{X}\hat{\beta}_1 = \hat{X}\hat{\beta}_2$ . II.  $\lambda'\hat{\beta}_1 = \lambda'\hat{\beta}_2$  for every estimable function  $\lambda'\beta$ . III.  $\hat{\beta}_1'X'Y = \hat{\beta}_2'X'Y$ , so the residual sum of squares is the same for both solutions. Which of the above statements are correct?

A. I and III only

B. II and III only

C. I and II only

D. I, II and III

Correct answer: D

## EXAM SHORTCUT

$\hat{X}\hat{\beta}$  is the (unique) projection of  $Y$  on  $\mathcal{C}(X)$ ; estimable  $\lambda' = a'X$  gives  $\lambda'\hat{\beta} = a'\hat{X}\hat{\beta}$ ;  $\text{SSE} = Y'Y - \hat{\beta}'X'Y$  is invariant.

Q6.

Let  $\lambda'\beta$  be estimable in  $Y = X\beta + \epsilon$ , where  $X$  is  $n \times p$  of rank  $k < p \leq n$ . Let  $\hat{\beta}$  be any solution of  $X'\hat{X}\hat{\beta} = X'Y$  and  $r$  any solution of  $X'Xr = \lambda$ . Consider the following statements: I.  $\lambda'\hat{\beta} = r'X'Y$  for any  $\hat{\beta}$  and any  $r$ . II. Both  $\lambda'\hat{\beta}$  and  $r'X'Y$  are invariant to the choices of  $\hat{\beta}$  and  $r$ . Which of the statements given above is/are correct?

A. Both I and II

B. I only

C. II only

D. Neither I nor II

Correct answer: A

## EXAM SHORTCUT

2026 Q59 archetype:  $\lambda'\hat{\beta} = r'X'\hat{X}\hat{\beta} = r'X'Y$ , and estimability makes both invariant.

Q7.

Consider the following statements: I. The column spaces satisfy  $\mathcal{C}(X'X) = \mathcal{C}(X')$ , which is why  $X'Y$  always lies in  $\mathcal{C}(X'X)$ . II.  $\text{rank}(X'X) = \text{rank}(X)$ . Which of the statements given above is/are correct?

- A. Neither I nor II
- B. I only
- C. Both I and II
- D. II only

Correct answer: C

## EXAM SHORTCUT

$\mathcal{C}(X'X) = \mathcal{C}(X')$  and equal ranks are the two standard lemmas behind consistency of the NE.

Q8.

If  $\hat{\beta}$  is any solution of the normal equations, the residual sum of squares equals

- A.  $\hat{\beta}'X'Y - Y'Y$
- B.  $Y'Y - \hat{\beta}'X'\hat{X}\hat{\beta} - \hat{\beta}'X'Y$
- C.  $Y'Y - \hat{\beta}'X'Y$
- D.  $Y'Y + \hat{\beta}'X'Y$

Correct answer: C

## EXAM SHORTCUT

$\text{SSE} = Y'Y - \hat{\beta}'X'Y$ ; the cross term collapses by the NE (2022 Q30 flavour).

Q9.

For the one-way model  $y_{ij} = \mu + \tau_i + e_{ij}$ ,  $i = 1, 2$ ;  $j = 1, 2$ , with group means  $\bar{y}_{1.}, \bar{y}_{2.}$  and grand mean  $\bar{y}_{..}$ , consider the following two candidate vectors  $(\hat{\mu}, \hat{\tau}_1, \hat{\tau}_2)$ : I.  $(0, \bar{y}_{1.}, \bar{y}_{2.})$  II.  $(\bar{y}_{..}, \bar{y}_{1.} - \bar{y}_{..}, \bar{y}_{2.} - \bar{y}_{..})$  Which of them solve the normal equations?

- A. I only
- B. Neither I nor II
- C. Both I and II
- D. II only

Correct answer: C

## EXAM SHORTCUT

Both satisfy  $\hat{\mu} + \hat{\tau}_i = \bar{y}_{i.}$ , hence both solve the NE — two different  $g$ -inverse choices; fitted values agree.

**Q10.**

In a less-than-full-rank Gauss–Markov model, consider the following statements: I.  $\hat{\beta}$  is not unique, yet the fitted vector  $\hat{Y} = \hat{X}\hat{\beta}$  is unique. II.  $E(\beta) = \beta$  for every solution  $\hat{\beta} = (X'X)^{-}X'Y$  of the normal equations. Which of the statements given above is/are correct?

- A. II only
- B. Neither I nor II
- C. **I only**
- D. Both I and II

**Correct answer: C**

**EXAM SHORTCUT**

$\hat{Y} = PY$  is the unique projection of  $Y$  on  $\mathcal{C}(X)$ ; but  $E(\beta) = (X'X)^{-}X'X\beta \neq \beta$  in general when  $\text{rank}(X) < p$ , so II fails — only estimable functions  $\lambda'\beta$  are unbiased.

# Hat Matrix, Fitted Values and Residuals

12 forecast questions in this topic

Linear Models

Hat Matrix, Fitted Values and Residuals

Medium

Q1.

In the full-rank model  $Y = X\beta + \epsilon$ , let  $H = X(X'X)^{-1}X'$ . Consider the following statements: 1.  $H$  is symmetric. 2.  $H$  is idempotent. 3.  $HX = X$ . Which of the above statements are correct?

A. 1 and 2 only

B. 2 and 3 only

C. 1, 2 and 3

D. 1 and 3 only

Correct answer: C

## EXAM SHORTCUT

$H' = H$ ,  $H^2 = H$ ,  $HX = X$ : the three defining projector facts.

Linear Models

Hat Matrix, Fitted Values and Residuals

Medium

Q2.

If  $X$  is  $n \times p$  of full column rank  $p$ , then  $\text{rank}(H)$  and  $\text{trace}(H)$  are respectively

A.  $p$  and  $p$

B.  $p$  and  $n-p$

C.  $n$  and  $n$

D.  $n-p$  and  $p$

Correct answer: A

## EXAM SHORTCUT

Idempotent:  $\text{rank} = \text{trace} = \text{trace}[(X'X)^{-1}X'X] = p$  (2019 Q2(a)).

Linear Models

Hat Matrix, Fitted Values and Residuals

Medium

Q3.

For a model containing an intercept (a column of ones), the leverages  $h_{ii}$  satisfy

A.  $h_{ii} = (p)/(n)$  for every  $i$

B.  $-1 \leq h_{ii} \leq 1$  and  $\sum_i h_{ii} = n-p$

C.  $0 \leq h_{ii} \leq (1)/(n)$

D.  $(1)/(n) \leq h_{ii} \leq 1$  and  $\sum_i h_{ii} = p$

Correct answer: D

## EXAM SHORTCUT

With intercept  $h_{ii} \geq 1/n$ ; always  $h_{ii} \leq 1$ ;  $\sum h_{ii} = \text{trace}(H) = p$ .

## Q4.

With  $\hat{Y} = HY$  and  $e = Y - \hat{Y}$ , and  $V(\epsilon) = \sigma^2 I$ , consider the following statements: 1.  $V(Y) = \sigma^2 H$ . 2.  $V(e) = \sigma^2(I - H)$ . 3.  $\text{Cov}(Y, e) = 0$ . Which of the above statements are correct?

- A. 2 and 3 only
- B. 1 and 2 only
- C. 1 and 3 only
- D. **1, 2 and 3**

Correct answer: D

## EXAM SHORTCUT

$V(HY) = \sigma^2 HH' = \sigma^2 H$ ;  $V((I - H)Y) = \sigma^2(I - H)$ ;  $\text{Cov} = \sigma^2 H(I - H) = 0$  (2019 Q21 archetype).

## Q5.

Let  $y \sim N(X\beta, \sigma^2 I)$  with  $X$  of full column rank  $p$ , and  $H = (h_{ij})$ . Which one of the following is not correct?

- A.  $\text{Cov}(e, \hat{Y}) = 0$
- B.  $\text{rank}(H) = p$
- C.  **$\text{Cov}(e_i, e_j) = \sigma^2 h_{ij}$  for  $i \neq j$**
- D.  $V(Y) = \sigma^2 H$

Correct answer: C

## EXAM SHORTCUT

For  $i \neq j$ ,  $\text{Cov}(e_i, e_j) = -\sigma^2 h_{ij}$ — the sign trap that decided 2019 Q2.

## Q6.

Consider the following statements: I.  $\text{Cov}(Y, \hat{Y}) = \sigma^2 H$ . II.  $V(e) = V(Y) - V(\hat{Y})$ . Which of the statements given above is/are correct?

- A. Neither I nor II
- B. II only
- C. I only
- D. **Both I and II**

Correct answer: D

## EXAM SHORTCUT

$\text{Cov}(Y, HY) = \sigma^2 H$ ; and  $\sigma^2(I - H) = \sigma^2 I - \sigma^2 H$ — the corrected version of 2019 Q21 statement 1 (which wrote + instead of − and was false).

Q7.

Consider the following statements about the residual vector  $e = (I-H)Y$ : 1.  $e = (I-H)\epsilon$ , so  $E(e) = 0$ . 2.  $X'e = 0$ : the residuals are orthogonal to every column of  $X$ . 3.  $\hat{Y}'e = 0$ . Which of the above statements are correct?

A. 1 and 2 only

B. **1, 2 and 3**

C. 1 and 3 only

D. 2 and 3 only

Correct answer: B

## EXAM SHORTCUT

$(I-H)X = 0$  kills the systematic part; orthogonality to  $\mathcal{C}(X)$  includes  $\hat{Y} = \hat{X}\beta$ .

Q8.

Consider the following statements: I. The residuals always sum to zero, whatever the design matrix. II. If the model contains an intercept term, then  $\sum e_i = 0$ . Which of the statements given above is/are correct?

A. I only

B. Both I and II

C. Neither I nor II

D. **II only**

Correct answer: D

## EXAM SHORTCUT

$\sum e_i = 1'e = 0$  requires  $1 \in \mathcal{C}(X)$ ; without an intercept it can fail.

Q9.

Consider the following statements about  $I-H$ : I.  $I-H$  is symmetric and idempotent with  $\text{rank}(I-H) = n-p$ . II. Every eigenvalue of  $H$  is either 0 or 1. Which of the statements given above is/are correct?

A. Neither I nor II

B. I only

C. **Both I and II**

D. II only

Correct answer: C

## EXAM SHORTCUT

$I-H$  projects on the error space of dimension  $n-p$ ; projector eigenvalues are 0/1.

## Q10.

The variance of the  $i$ -th residual is

- A.  $\sigma^2(1 + h_{ii})$
- B.  $\sigma^2$
- C.  $\sigma^2(1 - h_{ii})$
- D.  $\sigma^2 h_{ii}$

Correct answer: C

## EXAM SHORTCUT

Diagonal of  $\sigma^2(I - H)$ ; the basis of studentized residuals.

## Q11.

A model with an intercept and 3 regressors is fitted to  $n = 20$  observations. The average leverage  $h = (1/n) \sum h_{ii}$  equals

- A. 0.15
- B. 0.25
- C. 0.20
- D. 0.80

Correct answer: C

## EXAM SHORTCUT

$p = 4$  columns,  $\sum h_{ii} = 4$ ,  $h = 4/20 = 0.2$ .

## Q12.

Consider the following statements: I.  $He = 0$ . II.  $H\hat{Y} = \hat{Y}$ . Which of the statements given above is/are correct?

- A. II only
- B. Both I and II
- C. Neither I nor II
- D. I only

Correct answer: B

## EXAM SHORTCUT

$H(I - H) = 0$  and  $H^2 = H$ : fitted values are already in  $\mathcal{C}(X)$ .

# Variances and Covariances of Least-Squares Estimators

12 forecast questions in this topic

Linear Models

Variances and Covariances of Least-Squares Estimators

Medium

Q1.

In the simple linear regression model  $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ ,  $i = 1, \dots, n$ , with  $E(\epsilon_i) = 0$ ,  $V(\epsilon_i) = \sigma^2$ ,  $\text{Cov}(\epsilon_i, \epsilon_j) = 0$ , and  $S_{xx} = \sum (x_i - \bar{x})^2$ ,  $V(\beta_1)$  equals

A.  $(n\sigma^2)/(S_{xx})$

B.  $\sigma^2/(S_{xx})$

C.  $\sigma^2 S_{xx}$

D.  $(\sigma^2)/(n)$

Correct answer: B

## EXAM SHORTCUT

The 2023 Q60 formula:  $V(\beta_1) = \sigma^2/S_{xx}$ .

Linear Models

Variances and Covariances of Least-Squares Estimators

Medium

Q2.

In the model of F8.1,  $V(\beta_0)$  equals

A.  $(\sigma^2)/(S_{xx})$

B.  $\sigma^2 ((1)/(n) - \bar{x}^2/S_{xx})$

C.  $\sigma^2 ((1)/(n) + \bar{x}^2/S_{xx})$

D.  $(\sigma^2)/(n)$

Correct answer: C

## EXAM SHORTCUT

The 2023 Q58 formula:  $V(\beta_0) = \sigma^2(1/n + \bar{x}^2/S_{xx})$ .

Linear Models

Variances and Covariances of Least-Squares Estimators

Medium

Q3.

In the model of F8.1,  $\text{Cov}(\beta_0, \hat{\beta}_1)$  equals

A.  $\sigma^2 n S_{xx}$

B. 0 always

C.  $-\bar{x}\sigma^2/(S_{xx})$

D.  $\bar{x}\sigma^2/(S_{xx})$

Correct answer: C

## EXAM SHORTCUT

The 2023 Q59 formula:  $\text{Cov}(\beta_0, \hat{\beta}_1) = -\bar{x}\sigma^2/S_{xx}$ .

Q4.

Independent observations follow  $y_i = \beta_0 + \beta_1 x_i + u_i$ ,  $i = 1, \dots, 8$ , with error variance  $\sigma^2 = 2$ . If  $\sum x_i = 40$  and  $\sum x_i^2 = 240$ , then  $\text{Cov}(\hat{\beta}_0, \hat{\beta}_1)$  equals

- A. -2.5
- B. 0.25
- C. ~~-0.25~~
- D. 0

Correct answer: C

## EXAM SHORTCUT

$\bar{x} = 5$ ,  $S_{xx} = 240 - 8(25) = 40$ , so  $-\bar{x}\sigma^2/S_{xx} = -5(2)/40 = -0.25$  (2018 Q9 archetype, which gave -0.5).

Q5.

$\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = 0$  in simple linear regression if and only if

- A.  ~~$\bar{x} = 0$~~
- B.  $n$  is even
- C.  $\sigma^2 = 1$
- D.  $\bar{y} = 0$

Correct answer: A

## EXAM SHORTCUT

$\text{Cov} = -\bar{x}\sigma^2/S_{xx} = 0 \iff \bar{x} = 0$ .

Q6.

The variance of the estimated mean response  $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$  at  $x = x_0$  is  $\sigma^2 \left[ \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right]$ , which is minimized when

- A.  $x_0 = 0$
- B.  ~~$x_0 = \bar{x}$~~
- C.  $x_0 = \bar{y}$
- D.  $x_0$  is as large as possible

Correct answer: B

## EXAM SHORTCUT

Prediction is most precise at the centre of the design.

Q7.

Consider the following statements: I. The correlation between  $\hat{\beta}_0$  and  $\hat{\beta}_1$  has sign opposite to that of  $\bar{x}$ . II. Centring the regressor ( $x_i \mapsto x_i - \bar{x}$ ) makes  $\hat{\beta}_0$  and  $\hat{\beta}_1$  uncorrelated. Which of the statements given above is/are correct?

- A. I only
- B. **Both I and II**
- C. Neither I nor II
- D. II only

Correct answer: B

## EXAM SHORTCUT

$\text{Corr} \propto -\bar{x}$ ; centring gives  $\bar{x} = 0$ , an orthogonal design (links to S3).

Q8.

For  $n = 5$  observations with  $\sum x_i = 0$ ,  $\sum x_i^2 = 10$  and  $\sigma^2 = 2$ , the pair  $(V(\hat{\beta}_0), V(\hat{\beta}_1))$  equals

- A. (2, 10)
- B. (0.2, 0.4)
- C. (0.4, 0.4)
- D. **(0.4, 0.2)**

Correct answer: D

## EXAM SHORTCUT

$V(\hat{\beta}_0) = \sigma^2/n = 0.4$  (since  $\bar{x} = 0$ ),  $V(\hat{\beta}_1) = 2/10 = 0.2$ .

Q9.

If every regressor value is doubled ( $x_i \mapsto 2x_i$ ) while the responses are unchanged, consider the following statements: I. The new slope estimator is half the old one. II. The variance of the new slope estimator is one-fourth of the old one. Which of the statements given above is/are correct?

- A. I only
- B. Neither I nor II
- C. **Both I and II**
- D. II only

Correct answer: C

## EXAM SHORTCUT

$x \mapsto cx: \hat{\beta}_1 \mapsto \hat{\beta}_1/c$  and  $V = \sigma^2/(c^2 S_{xx})$ ; with  $c = 2$ , half and one-fourth.

**Q10.**

Other things being equal,  $V(\beta_1)$  decreases when

- A. the responses  $y_i$  are centred about  $\bar{y}$
- B. the  $x_i$  are chosen closer to  $\bar{x}$
- C.  $\sigma^2$  increases
- D. the spread  $S_{xx}$  of the regressor values increases

**Correct answer: D**

**EXAM SHORTCUT**

$V(\beta_1) = \sigma^2 / S_{xx}$ : spreading the design points inflates  $S_{xx}$  and shrinks the variance; (b) and (c) inflate it, (a) changes nothing.

**Q11.**

In the full-rank model  $Y = X\beta + \epsilon$ ,  $V(\epsilon) = \sigma^2 I$ , consider the following statements: I.  $V(\beta) = \sigma^2 (X'X)^{-1}$ . II.  $\text{Cov}(\lambda'\beta, \mu'\beta) = \sigma^2 \lambda'(X'X)^{-1}\mu$ . Which of the statements given above is/are correct?

- A. Both I and II
- B. II only
- C. I only
- D. Neither I nor II

**Correct answer: A**

**EXAM SHORTCUT**

Both are the standard full-rank dispersion facts (2019 Q25 archetype).

**Q12.**

Let  $\lambda'\beta$  be estimable in a (possibly less-than-full-rank) Gauss–Markov model. Consider the following statements: I.  $V(\lambda'\beta) = \sigma^2 \lambda'(X'X)^{-}\lambda$ . II. The value  $\lambda'(X'X)^{-}\lambda$  is the same for every generalized inverse  $(X'X)^{-}$ . Which of the statements given above is/are correct?

- A. Both I and II
- B. I only
- C. Neither I nor II
- D. II only

**Correct answer: A**

**EXAM SHORTCUT**

For estimable  $\lambda'\beta$ ,  $\lambda \in \mathcal{C}(X'X)$  makes the quadratic form invariant; it is the BLUE's variance.

# Estimation with Correlated Observations: GLS, WLS, Aitken

12 forecast questions in this topic

Linear Models

Estimation with Correlated Observations: GLS, WLS, Aitken

Medium

Q1.

In the model  $Y = X\beta + \epsilon$  with  $E(\epsilon) = 0$  and  $V(\epsilon) = \sigma^2 V$ ,  $V$  known and positive definite, the generalized least-squares (Aitken) estimator of  $\beta$  is

- A.  $V^{-1}(X'X)^{-1}X'Y$
- B.  $(X'X)^{-1}X'Y$
- C.  $(X'VX)^{-1}X'VY$
- D.  $(X'V^{-1}X)^{-1}X'V^{-1}Y$

Correct answer: D

## EXAM SHORTCUT

Aitken estimator: transform by  $V^{-1/2}$  and apply OLS.

Linear Models

Estimation with Correlated Observations: GLS, WLS, Aitken

Medium

Q2.

The correlated model  $E(Y) = X\beta$ ,  $D(Y) = \sigma^2 G$  with  $|G| \neq 0$  is reduced to a standard Gauss–Markov model by the transformation

- A.  $Z = G^{-1/2}Y$
- B.  $Z = G^{-1}Y$
- C.  $Z = G^{1/2}Y$
- D.  $Z = Y - GY$

Correct answer: A

## EXAM SHORTCUT

$D(G^{-1/2}Y) = \sigma^2 G^{-1/2}GG^{-1/2} = \sigma^2 I$  (2025 Q58 verbatim archetype).

Linear Models

Estimation with Correlated Observations: GLS, WLS, Aitken

Medium

Q3.

When  $V(\epsilon) = \sigma^2 V$  with  $V \neq I$ , consider the following statements about the ordinary least-squares estimator  $\hat{\beta}_{OLS} = (X'X)^{-1}X'Y$ : I.  $\hat{\beta}_{OLS}$  remains unbiased for  $\beta$ . II.  $\hat{\beta}_{OLS}$  remains the best linear unbiased estimator of  $\beta$  in general. Which of the statements given above is/are correct?

- A. Both I and II
- B. Neither I nor II
- C. II only
- D. I only

Correct answer: D

## EXAM SHORTCUT

Unbiasedness needs only  $E(\epsilon) = 0$ ; efficiency is lost — GLS is BLUE.

Q4.

Aitken's theorem states that under  $E(\epsilon) = 0$ ,  $V(\epsilon) = \sigma^2 V$  with  $V$  known positive definite, the BLUE of an estimable  $\lambda'\beta$  is obtained from

- A. **generalized least squares, i.e., minimizing  $(Y - X\beta)'V^{-1}(Y - X\beta)$**
- B. minimizing  $(Y - X\beta)'V(Y - X\beta)$
- C. ordinary least squares applied to  $Y$
- D. maximum likelihood, which requires normality

Correct answer: A

**EXAM SHORTCUT**

Weight by the inverse dispersion; no normality needed.

Q5.

Let  $y_1, y_2, y_3$  be independent with common mean  $\mu$  and variances  $V(y_i) = \sigma^2/w_i$ , where  $(w_1, w_2, w_3) = (1, 2, 3)$ . The BLUE of  $\mu$  is

- A.  $(y_1 + 4y_2 + 9y_3)/(14)$
- B.  **$(y_1 + 2y_2 + 3y_3)/(6)$**
- C.  $(3y_1 + 2y_2 + y_3)/(6)$
- D.  $(y_1 + y_2 + y_3)/(3)$

Correct answer: B

**EXAM SHORTCUT**

$\hat{\mu} = \sum w_i y_i / \sum w_i = (y_1 + 2y_2 + 3y_3)/6$  (2025 Q63 archetype in weighted-mean form).

Q6.

In the setting of F9.5, the variance of the BLUE of  $\mu$  is

- A.  $(\sigma^2)/(14)$
- B.  $(\sigma^2)/(3)$
- C.  $(11\sigma^2)/(36)$
- D.  **$(\sigma^2)/(6)$**

Correct answer: D

**EXAM SHORTCUT**

$V(\hat{\mu}) = \sigma^2 / \sum w_i = \sigma^2/6$  (2025 Q64 archetype).

Q7.

Consider the following statements: I. If  $V = I$ , the GLS estimator reduces to the OLS estimator. II. OLS and GLS coincide whenever  $\mathcal{C}(VX) \subseteq \mathcal{C}(X)$ . Which of the statements given above is/are correct?

- A. I only
- B. ~~A~~Both I and II
- C. Neither I nor II
- D. II only

Correct answer: B

## EXAM SHORTCUT

$V = I$  is trivial; the general coincidence condition is  $\mathcal{C}(VX) \subseteq \mathcal{C}(X)$  (equivalently  $VX = XQ$ ).

Q8.

Let  $y_1, \dots, y_n$  be equicorrelated:  $V(y_i) = \sigma^2$ ,  $\text{Corr}(y_i, y_j) = \rho$  for  $i \neq j$ . For the dispersion matrix to be positive definite,  $\rho$  must satisfy

- A.  $-1/n < \rho < 1/n$
- B.  $-1 < \rho < 1$
- C. ~~A~~ $-1/(n-1) < \rho < 1$
- D.  $0 \leq \rho < 1$

Correct answer: C

## EXAM SHORTCUT

Eigenvalues of  $(1-\rho)I + \rho J$  are  $1 + (n-1)\rho$  and  $1-\rho$ : both positive iff  $-1/(n-1) < \rho < 1$ .

Q9.

In the equicorrelated model of F9.8 with common mean  $E(y_i) = \mu$ , consider the following statements: I.  $\bar{y}$  is the BLUE of  $\mu$ . II.  $V(\bar{y}) = (\sigma^2)/(n)[1 + (n-1)\rho]$ . Which of the statements given above is/are correct?

- A. I only
- B. II only
- C. Neither I nor II
- D. ~~A~~Both I and II

Correct answer: D

## EXAM SHORTCUT

$V_1 = [1 + (n-1)\rho]1 \in \mathcal{C}(X)$ , so OLS = GLS and  $\bar{y}$  stays BLUE; its variance carries the correlation inflation factor.

## Q10.

With heteroscedastic independent errors,  $V(y_i) = \sigma^2/w_i$ , weighted least squares chooses  $\beta$  to minimize

- A.  $\sum_i w_i^2(y_i - x_i'\beta)^2$
- B.  $\sum_i (y_i - x_i'\beta)^2/w_i$
- C.  $\sum_i (y_i - x_i'\beta)^2$
- D.  $\sum_i w_i(y_i - x_i'\beta)^2$

Correct answer: D

## EXAM SHORTCUT

Weights are inverse variances: minimize  $\sum w_i(y_i - x_i'\beta)^2$ .

## Q11.

Consider the following statements: I.  $V(\beta_{GLS}) = \sigma^2(X'V^{-1}X)^{-1}$ . II.  $V(\beta_{OLS}) - V(\beta_{GLS})$  is non-negative definite. Which of the statements given above is/are correct?

- A. Neither I nor II
- B. Both I and II
- C. I only
- D. II only

Correct answer: B

## EXAM SHORTCUT

Standard GLS dispersion and the Gauss–Markov efficiency ordering.

## Q12.

Five equicorrelated observations have  $\sigma^2 = 10$  and  $\rho = 0.2$ , with common mean  $\mu$ . Then  $V(\bar{y})$  equals

- A. 3.6
- B. 5.0
- C. 2.0
- D. 1.8

Correct answer: A

## EXAM SHORTCUT

$(10)/(5) [1 + 4(0.2)] = 2 \times 1.8 = 3.6$ .

# The Gauss–Markov Theorem and Properties of LSEs

12 forecast questions in this topic

Linear Models

The Gauss–Markov Theorem and Properties of LSEs

Medium

**Q1.**

Consider the following statements about the Gauss–Markov theorem: I. It requires the errors to be normally distributed. II. It requires only  $E(\epsilon) = 0$  and  $V(\epsilon) = \sigma^2 I$  (zero-mean, homoscedastic, uncorrelated errors). Which of the statements given above is/are correct?

A. Neither I nor II

B. I only

**C. ~~II~~ only**

D. Both I and II

**Correct answer: C**

## EXAM SHORTCUT

No normality anywhere in Gauss–Markov: second-moment assumptions only.

Linear Models

The Gauss–Markov Theorem and Properties of LSEs

Medium

**Q2.**

The Gauss–Markov theorem asserts that for any estimable function  $\lambda'\beta$ , the least-squares estimator  $\lambda'\hat{\beta}$

A. is unbiased only under normality

B. has minimum mean squared error among all linear estimators

C. has minimum variance among all unbiased estimators

**D. ~~has~~ minimum variance in the class of linear unbiased estimators of  $\lambda'\beta$**

**Correct answer: D**

## EXAM SHORTCUT

“Best” = minimum variance within the linear unbiased class; nothing is claimed about non-linear or biased competitors.

Linear Models

The Gauss–Markov Theorem and Properties of LSEs

Medium

**Q3.**

For the full-rank model  $y = X\beta + u$ ,  $E(u) = 0$ ,  $V(u) = \sigma^2 I$ ,  $\hat{\beta} = (X'X)^{-1}X'y$ , which of the following statements are correct? 1.  $V(\hat{\beta}) = \sigma^2(X'X)^{-1}$ . 2.  $\hat{\beta}$  is the BLUE of  $\beta$ . 3. The residual mean square is an unbiased estimator of  $\sigma^2$ . Select the correct answer using the code given below.

A. 1 and 2 only

**B. ~~1~~, 2 and 3**

C. 2 and 3 only

D. 1 and 3 only

**Correct answer: B**

## EXAM SHORTCUT

2019 Q25 archetype: all three are correct.

Q4.

Consider the following statements: I. Even when the normal equations have infinitely many solutions  $\hat{\beta}$ , the least-squares estimator of an estimable function is unique. II. The BLUE of an estimable function is unique. Which of the statements given above is/are correct?

- A. II only
- B. I only
- C. Both I and II
- D. Neither I nor II

Correct answer: C

## EXAM SHORTCUT

$\lambda'\hat{\beta}$  is invariant across NE solutions and is the (a.s. unique) BLUE.

Q5.

In the model  $y_i = \alpha + \beta x_i + u_i$ , both  $x$  and  $y$  are measured in metres. If all measurements are converted to centimetres (both variables multiplied by 100), then

- A.  $\hat{\beta}$  changes but  $\hat{\alpha}$  does not change
- B. neither  $\hat{\alpha}$  nor  $\hat{\beta}$  changes
- C.  $\hat{\alpha}$  changes but  $\hat{\beta}$  does not change
- D. both  $\hat{\alpha}$  and  $\hat{\beta}$  change

Correct answer: C

## EXAM SHORTCUT

2019 Q26 archetype: common rescaling  $y, x \rightarrow cy, cx$  leaves  $\hat{\beta} = S_{xy}/S_{xx}$  unchanged but multiplies  $\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$  by  $c$ .

Q6.

Models I:  $y = \beta_0 + \beta_1 x$  and II:  $y = \beta_0 + \beta_1 x + \beta_2 x^2$  are fitted by least squares to the same data, with residual sums of squares  $P$  and  $Q$  respectively. Consider the following statements: I.  $P \geq Q$ . II. It can happen that  $Q = 0$  but  $P > 0$ . Which of the statements given above is/are correct?

- A. Neither I nor II
- B. Both I and II
- C. II only
- D. I only

Correct answer: B

## EXAM SHORTCUT

2019 Q3 = 2026 Q57: enlarging the model never increases SSE; a quadratic can interpolate data that a line cannot.

Q7.

For the same data set, Student-I fits  $\hat{Y}_i = \alpha_0 + \alpha_1 X_i$  ( $Y$  on  $X$ ) and Student-II fits  $\hat{X}_i = \beta_0 + \beta_1 Y_i$  ( $X$  on  $Y$ ) by ordinary least squares. Which one of the following is a possible value of  $(\alpha_1, \beta_1)$ ?

- A. (0.9, 1.25)
- B. (1.5, 0.9)
- C. ~~(0.8, 1.2)~~
- D. (−0.6, 0.8)

**Correct answer: C**

#### EXAM SHORTCUT

Need same sign and product  $\leq 1$ :  $(0.8)(1.2) = 0.96 \leq 1$ ; (a) gives  $1.125 > 1$ , (d) mixes signs, (b) gives  $1.35 > 1$  (2026 Q58 archetype).

Q8.

Consider the following statements about the two OLS slopes of F10.7: I.  $\hat{\alpha}_1 \hat{\beta}_1 = r^2$ , the squared sample correlation coefficient. II. The two slopes always have the same sign. Which of the statements given above is/are correct?

- A. I only
- B. ~~Both I and II~~
- C. Neither I nor II
- D. II only

**Correct answer: B**

#### EXAM SHORTCUT

$\hat{\alpha}_1 \hat{\beta}_1 = (S_{xy}/S_{xx}) \cdot (S_{xy}/S_{yy}) = r^2 \leq 1$ , and both slopes carry the sign of  $S_{xy}$ .

Q9.

Consider the following statements: I. Under perfect (exact) multicollinearity,  $(X'X)^{-1}$  does not exist. II. The problem of (near) multicollinearity can be mitigated by increasing the sample size or collecting additional data. Which of the statements given above is/are correct?

- A. I only
- B. II only
- C. Neither I nor II
- D. ~~Both I and II~~

**Correct answer: D**

#### EXAM SHORTCUT

2018 Q3/Q4 archetypes: exact collinearity kills the inverse; more data can break near-collinearity.

## Q10.

Consider the following statements: I. The method of least squares is an algebraic (mathematical) fitting method requiring no distributional assumptions. II. Under normal errors, the least-squares estimator of  $\beta$  coincides with the maximum likelihood estimator. Which of the statements given above is/are correct?

- A. I only
- B. Both I and II
- C. II only
- D. Neither I nor II

Correct answer: B

## EXAM SHORTCUT

Least squares is distribution-free algebra (the 2018 Q3 trap: it is not a “statistical” method); under normality LSE = MLE for  $\beta$ .

## Q11.

For the Gauss–Markov model  $E(y_i) = \alpha_1 \beta_1 + \dots + \alpha_p \beta_p$ ,  $V(y_i) = \sigma^2$ ,  $\text{cov}(y_i, y_j) = 0$ , consider the following statements: 1. If the  $p$  columns  $\alpha_1, \dots, \alpha_p$  are linearly independent ( $p \leq n$ ), the model is of full rank. 2. If all the  $\beta$ 's are fixed constants, the model is a fixed-effects (Model I) model. Which of the above statements is/are correct?

- A. Neither 1 nor 2
- B. 2 only
- C. 1 only
- D. Both 1 and 2

Correct answer: D

## EXAM SHORTCUT

2019 Q30 verbatim archetype: both definitions are correct.

## Q12.

Consider the following statements: I. Unbiasedness of  $\hat{\beta}$  (full-rank case) requires only  $E(\epsilon) = 0$ . II. Unbiasedness of  $\hat{\sigma}^2 = \text{SSE}/(n-p)$  for  $\sigma^2$  requires normality of the errors. Which of the statements given above is/are correct?

- A. Both I and II
- B. Neither I nor II
- C. I only
- D. II only

Correct answer: C

## EXAM SHORTCUT

$E(\text{SSE}) = (n-p)\sigma^2$  follows from trace algebra alone — no normality needed; normality is needed only for the  $\chi^2$  distribution of SSE.

# Error Space and Estimation Space

10 forecast questions in this topic

Linear Models

Error Space and Estimation Space

Medium

Q1.

In the Gauss–Markov model  $Y_{n \times 1} = X_{n \times k} \beta + \epsilon$  with  $\text{rank}(X) = r < k$ , the set  $\{l \in \mathbb{R}^n : E(l'Y) = 0 \text{ for all } \beta\}$  is

- A. not a subspace of  $\mathbb{R}^n$
- B. **a subspace of dimension  $n-r$**
- C. a subspace of dimension  $n-k$
- D. a subspace of dimension  $k-r$

Correct answer: B

## EXAM SHORTCUT

$E(l'Y) = l'X\beta = 0 \Leftrightarrow X'l = 0$ : the null space of  $X'$  has dimension  $n-r$  (2026 Q60 verbatim archetype).

Linear Models

Error Space and Estimation Space

Medium

Q2.

Consider the following statements: 1. The estimation space is the column space  $\mathcal{C}(X)$ , of dimension  $r$ . 2. The error space is the orthogonal complement  $\mathcal{C}(X)^\perp$ , of dimension  $n-r$ . 3. Every vector of the error space is orthogonal to every vector of the estimation space. Which of the above statements are correct?

- A. **1, 2 and 3**
- B. 1 and 3 only
- C. 1 and 2 only
- D. 2 and 3 only

Correct answer: A

## EXAM SHORTCUT

The two spaces are orthogonal complements with dimensions  $r$  and  $n-r$ .

Linear Models

Error Space and Estimation Space

Medium

Q3.

A linear function  $l'Y$  is the BLUE of its own expectation if and only if

- A.  **$l$  lies in the estimation space  $\mathcal{C}(X)$**
- B.  $l' = 1$
- C.  $l$  lies in the error space
- D.  $l$  has all components equal

Correct answer: A

## EXAM SHORTCUT

BLUES are exactly the linear functions with coefficient vector in  $\mathcal{C}(X)$ .

Q4.

Consider the following statements about error functions (linear functions  $z'Y$  with  $z$  in the error space): I.  $E(z'Y) = 0$  identically in  $\beta$ . II. Error functions are the raw material for estimating  $\sigma^2$ . Which of the statements given above is/are correct?

- A. Neither I nor II
- B. I only
- C. Both I and II
- D. II only

Correct answer: C

## EXAM SHORTCUT

Zero expectation for all  $\beta$ ; squares of error functions estimate  $\sigma^2$  (S4 archetype machinery).

Q5.

A Gauss–Markov model has  $n = 7$  observations and  $\text{rank}(X) = 3$ . The dimensions of the estimation space and the error space are respectively

- A. 3 and 7
- B. 3 and 4
- C. 7 and 3
- D. 4 and 3

Correct answer: B

## EXAM SHORTCUT

Dimensions  $r = 3$  and  $n - r = 4$ .

Q6.

Every vector  $a \in \mathbb{R}^n$  decomposes uniquely as  $a = a_1 + a_2$  with  $a_1 \in \mathcal{C}(X)$  and  $a_2 \in \mathcal{C}(X)^\perp$ . The BLUE of  $E(a'Y)$  is then

- A.  $a'Y$
- B.  $a_1'Y$
- C.  $a_1'Y$
- D.  $(a_1 - a_2)'Y$

Correct answer: C

## EXAM SHORTCUT

Project  $a$  onto  $\mathcal{C}(X)$ :  $E(a'Y) = E(a_1'Y)$  and  $a_1'Y$  is BLUE with smaller variance ( $\|a_1\| \leq \|a\|$ ).

Q7.

If  $l \in \mathcal{C}(X)$  and  $z$  is in the error space, then under  $V(\epsilon) = \sigma^2 I$ ,  $\text{Cov}(l'Y, z'Y)$  equals

A.  $\sigma^2 l'z = 0$ B.  $\sigma^2(l'l)(z'z)$ C.  $\sigma^2$ 

D. cannot be determined

**Correct answer: A****EXAM SHORTCUT**

$\text{Cov} = \sigma^2 l'z$  and orthogonality gives  $l'z = 0$ .

Q8.

Consider the following characterization: a linear unbiased estimator  $l'Y$  of  $\lambda'\beta$  is the BLUE if and only if

A. its variance is  $\sigma^2$ B.  $l$  is a unit vector

C. it is correlated with every error function

D. **It is uncorrelated with every error function****Correct answer: D****EXAM SHORTCUT**

The classical characterization: zero covariance with every zero-expectation linear function.

Q9.

With  $n$  observations and  $\text{rank}(X) = r$ , consider the following statements: I. The maximum number of linearly independent error functions is  $n-r$ . II. The maximum number of linearly independent BLUEs (equivalently, of linearly independent estimable functions) is  $r$ . Which of the statements given above is/are correct?

A. I only

B. **Both I and II**

C. Neither I nor II

D. II only

**Correct answer: B****EXAM SHORTCUT**

Bases of the two orthogonal complements.

**Q10.**

Consider the following statements: 1. The fitted vector  $\hat{Y}$  lies in the estimation space. 2. The residual vector  $e$  lies in the error space. 3.  $\hat{Y}'e = 0$ . Which of the above statements are correct?

- A. 1 and 3 only
- B. 1 and 2 only
- C. 1, 2 and 3
- D. 2 and 3 only

**Correct answer: C**

**EXAM SHORTCUT**

$\hat{Y} = PY \in \mathcal{C}(X)$ ,  $e = (I - P)Y \perp \mathcal{C}(X)$ .

# Distribution Theory and Quadratic Forms

12 forecast questions in this topic

Linear Models

Distribution Theory and Quadratic Forms

Medium

Q1.

Let  $Y \sim N_n(\mu, \Sigma)$  with  $\Sigma$  positive definite. Then  $E(Y'AY)$  equals

A.  $\text{trace}(A\Sigma) + \mu' A \mu$

B.  $\text{trace}(A) + \mu' \Sigma \mu$

C.  $\text{trace}(A\Sigma)$

D.  $\mu' A \mu$

Correct answer: A

## EXAM SHORTCUT

The fundamental expectation identity for quadratic forms.

Linear Models

Distribution Theory and Quadratic Forms

Medium

Q2.

Let  $Y \sim N_3(0, I_3)$  and  $A = \text{diag}(1, 1, 0)$ . Then  $\text{Var}(Y'AY)$  equals

A. 6

B. 1

C. 4

D. 2

Correct answer: C

## EXAM SHORTCUT

For  $N(0, I)$ :  $\text{Var}(Y'AY) = 2 \text{trace}(A^2) = 2(2) = 4$  (2023 Q53 archetype, which had  $\text{trace}(A^2) = 1$ , answer 2).

Linear Models

Distribution Theory and Quadratic Forms

Medium

Q3.

Let  $Y \sim N_3(\mu, I_3)$  with  $\mu = (1, 2, 1)'$ . Then  $E(Y'Y)$  equals

A. 3

B. 12

C. 6

D. 9

Correct answer: D

## EXAM SHORTCUT

$\text{trace}(I_3) + \mu' \mu = 3 + (1 + 4 + 1) = 9$ .

Q4.

Let  $Y \sim N_3(\mu, \Sigma)$  with  $\mu' = (2, -1, 3)$  and  $\Sigma = \text{diag}(4, 1, 9)$ . Then  $Y'\Sigma^{-1}Y$  follows a non-central chi-square distribution with 3 degrees of freedom and non-centrality parameter  $\lambda = \mu'\Sigma^{-1}\mu$  equal to

- A. 14
- B. 1.5
- C. 3
- D. 6

Correct answer: C

## EXAM SHORTCUT

$\lambda = 4/4 + 1/1 + 9/9 = 3$  (2023 Q51 archetype with the  $\lambda = \mu'\Sigma^{-1}\mu$  convention stated explicitly).

Q5.

Let  $Y \sim N_n(0, \Sigma)$ . Consider the following statements: I.  $Y'AY \sim \chi^2$  if and only if  $A\Sigma$  is idempotent. II. When it holds, the degrees of freedom equal  $\text{rank}(A)$ . Which of the statements given above is/are correct?

- A. I only
- B. Both I and II
- C. Neither I nor II
- D. II only

Correct answer: B

## EXAM SHORTCUT

The classical  $\chi^2$  criterion for quadratic forms.

Q6.

Let  $Y \sim N_n(\mu, \sigma^2 I)$ . The quadratic forms  $Y'AY$  and  $Y'BY$  are independently distributed if and only if

- A.  $A$  and  $B$  are both singular
- B.  $\text{trace}(AB) = 1$
- C.  $AB = 0$
- D.  $A + B = I$

Correct answer: C

## EXAM SHORTCUT

Craig's theorem (sufficient version used at exam level):  $AB = 0$ .

Q7.

Under the normal full-rank model, consider the following statements: I.  $\hat{\beta}$  and SSE are independently distributed. II. This follows because  $(X'X)^{-1}X' \cdot (I-H) = 0$ . Which of the statements given above is/are correct?

A. Both I and II

B. II only

C. I only

D. Neither I nor II

Correct answer: A

## EXAM SHORTCUT

Linear form and quadratic form are independent since the product of their matrices vanishes:  $X'(I-H) = 0$ .

Q8.

(Cochran's theorem) Let  $Y \sim N_n(0, I)$  and  $Y'Y = Q_1 + \dots + Q_k$  where each  $Q_j = Y'A_jY$  has  $\text{rank}(A_j) = n_j$ . Then  $Q_1, \dots, Q_k$  are independent  $\chi^2_{n_j}$  variables if and only if

A.  $\sum n_j = n^2$ B.  $\sum n_j = n$ C.  $k = 2$ D. each  $n_j = 1$ 

Correct answer: B

## EXAM SHORTCUT

Rank additivity  $\sum n_j = n$  is exactly Cochran's condition.

Q9.

Let RSS be the unconditional residual sum of squares and RSS1 the residual sum of squares under  $m$  independent estimable restrictions  $l_i'\beta = \zeta_i$  (rank of  $X$  is  $r$ ). Under normality, consider the following statements: 1. When the restrictions are true,  $(\text{RSS1} - \text{RSS})/\sigma^2 \sim \chi^2_m$ . 2.  $\text{RSS1}/\text{RSS} \sim F_{\{m, n-r\}}$ . 3.  $\text{RSS1} - \text{RSS}$  and  $\text{RSS}$  are independently distributed. Which of the above statements are correct?

A. 1 and 2 only

B. 2 and 3 only

C. 1 and 3 only

D. 1, 2 and 3

Correct answer: C

## EXAM SHORTCUT

2019 Q22 archetype, corrected: the F ratio is  $[(\text{RSS1}-\text{RSS})/m] / [\text{RSS}/(n-r)]$ , not  $\text{RSS1}/\text{RSS}$ ; 1 and 3 hold.

**Q10.**

Under the normal Gauss–Markov model with  $\text{rank}(X) = r$  and estimable  $\lambda'\beta$ , the statistic  $(\lambda'\hat{\beta} - \lambda'\beta) / \sqrt{(\sigma^2 \lambda'(X'X)^{-1}\lambda)}$ , with  $\sigma^2 = \text{SSE}/(n-r)$ , follows

- A.  $tn$
- B.  $N(0, 1)$
- C.  $\sqrt{n-r}$
- D.  $\chi^2_{n-r}$

**Correct answer: C**

**EXAM SHORTCUT**

Normal numerator over independent  $\sqrt{\chi^2_{n-r}/(n-r)}$ .

**Q11.**

Under normality, consider the following statements: I.  $\hat{Y}$  and  $e$  are jointly normal and uncorrelated, hence independent. II. For jointly normal vectors, zero covariance implies independence. Which of the statements given above is/are correct?

- A. II only
- B. Neither I nor II
- C. Both I and II
- D. I only

**Correct answer: C**

**EXAM SHORTCUT**

$\text{Cov}(Y, e) = 0$  plus joint normality.

**Q12.**

In the normal full-rank model, the exact distribution of  $\hat{\beta}$  is

- A.  $N(\beta, \sigma^2 I)$
- B.  $tn-p$
- C.  $N(0, \sigma^2(X'X)^{-1})$
- D.  $N(\beta, \sigma^2(X'X)^{-1})$

**Correct answer: D**

**EXAM SHORTCUT**

Linear transform of a normal vector; mean  $\beta$ , dispersion  $\sigma^2(X'X)^{-1}$ .

# One-Way Classification

12 forecast questions in this topic

Linear Models

One-Way Classification

Medium

**Q1.**

For the one-way model  $y_{ij} = \mu + \alpha_i + e_{ij}$ ,  $i = 1, \dots, k$ ;  $j = 1, \dots, n_i$ , consider the following statements: I. The design matrix has  $k + 1$  columns but rank  $k$ . II.  $\mu + \alpha_i$  and every contrast  $\sum j_i \alpha_i$  ( $\sum j_i = 0$ ) are estimable, but  $\mu$  alone is not. Which of the statements given above is/are correct?

A. I only

**B. Both I and II**

C. II only

D. Neither I nor II

**Correct answer: B**

## EXAM SHORTCUT

Column of ones = sum of the indicator columns: rank  $k$ ; the estimable functions are spanned by the cell means  $\mu + \alpha_i$ .

Linear Models

One-Way Classification

Medium

**Q2.**

Six treatments are each replicated 4 times in a completely randomized experiment. The degrees of freedom for treatments and error are respectively

A. 5 and 23

**B. 5 and 18**

C. 6 and 17

D. 6 and 18

**Correct answer: B**

## EXAM SHORTCUT

$k - 1 = 5$ ,  $N - k = 24 - 6 = 18$  (2021 Q8 archetype: there  $30 - 5 = 25$ ).

Linear Models

One-Way Classification

Medium

**Q3.**

In one-way ANOVA with  $k$  groups and  $n$  observations in all, consider the following statements: I. The test statistic is  $F = \text{MST}/\text{MSE}$ . II. Under  $H_0: \alpha_1 = \dots = \alpha_k$ ,  $F \sim F_{k-1, n-k}$ . Which of the statements given above is/are correct?

A. Neither I nor II

B. II only

**C. Both I and II**

D. I only

**Correct answer: C**

## EXAM SHORTCUT

Definition of the variance-ratio test (2022 Q6 archetype).

Q4.

In the balanced fixed-effects one-way model ( $n_i = n$ ,  $\sum i a_i = 0$ ),  $E(\text{MST})$  equals

A.  $\sigma^2 + n\sigma^2_{\alpha}/(k-1)$ B.  $\sigma^2$  alwaysC.  $\sigma^2 + n\sigma^2_{\alpha}$ D.  $\sigma^2/n$ 

Correct answer: A

## EXAM SHORTCUT

Fixed effects:  $E(\text{MST}) = \sigma^2 + n\sigma^2_{\alpha}/(k-1)$ ; (c) is the random-effects line — the classic confusion pair.

Q5.

For the cell-means model  $E(Y_{ij}) = \mu_i$ ,  $\text{var}(Y_{ij}) = \sigma^2$ ,  $j = 1, \dots, n_i$ ,  $i = 1, \dots, p$ , consider the following statements:

1. Every linear parametric function  $\sum j c_j \mu_i$  is estimable. 2.  $\hat{\mu}_i = \bar{Y}_i = n_i^{-1} \sum j Y_{ij}$ . Which of the above statements is/are correct?

A. 1 only

B. Both 1 and 2

C. Neither 1 nor 2

D. 2 only

Correct answer: B

## EXAM SHORTCUT

2018 Q13 verbatim archetype: full-rank cell-means model, so everything is estimable and group means are the LSEs.

Q6.

Weights of fish are sampled from three lakes and a one-way ANOVA is performed. The null hypothesis of the analysis is

A.  $H_0$ : at least one pair of lakes has equal mean weightsB.  $H_0: \mu_1 = \mu_2 = \mu_3 (= \mu)$ C.  $H_0: \mu_1 = \mu_2 = \mu_3 = 0$ D.  $H_0: \sigma^2_1 = \sigma^2_2 = \sigma^2_3$ 

Correct answer: B

## EXAM SHORTCUT

Equality of means, not variances, and not “= 0” (2020 Q41 archetype).

Q7.

If the computed  $F$  exceeds the critical value at the chosen level, the correct conclusion of the one-way ANOVA is that

- A. **At least one population mean differs from the others**
- B. the population variances are unequal
- C. all the population means are different from one another
- D. exactly two of the population means differ

Correct answer: A

#### EXAM SHORTCUT

Rejection says at least one differs — never “all differ” (2020 Q42 archetype).

Q8.

In a one-way layout with  $k = 3$  groups and  $N = 15$  observations in all, the total sum of squares is 400 and the between-groups (treatment) sum of squares is 160. The value of the  $F$  statistic is

- A. 8
- B. **4**
- C. 20
- D. 2

Correct answer: B

#### EXAM SHORTCUT

$SSE = 240$ ;  $F = (160/2)/(240/12) = 80/20 = 4$ .

Q9.

Consider the following statements about the side condition  $\sum i n_i a_i = 0$  (or  $\sum i a_i = 0$ ) in the one-way model: I. It is an identifiability constraint that selects one particular solution of the normal equations. II. Imposing it changes the class of estimable functions. Which of the statements given above is/are correct?

- A. II only
- B. Both I and II
- C. **I only**
- D. Neither I nor II

Correct answer: C

#### EXAM SHORTCUT

Constraints pick a solution; estimability is a property of the model, not of the constraint.

Q10.

With  $k$  groups and  $n$  total observations, the one-way ANOVA identity partitions the total degrees of freedom as

- A.  $n = (k) + (n-k)$
- B.  $n-1 = (k-1) + (n-k)$
- C.  $n+1 = (k+1) + (n-k)$
- D.  $n-1 = k + (n-k-1)$

Correct answer: B

## EXAM SHORTCUT

TSS df =  $n-1$  splits into  $k-1$  and  $n-k$ .

Q11.

For a contrast  $\psi = \sum j c_i \mu_i$  ( $\sum j c_i = 0$ ) estimated by  $\hat{\psi} = \sum j c_i \bar{y}_i$ , consider the following statements: I.  $E(\hat{\psi}) = \psi$ . II.  $V(\hat{\psi}) = \sigma^2 \sum j c_i^2 / n_i$ . Which of the statements given above is/are correct?

- A. II only
- B. Neither I nor II
- C. Both I and II
- D. I only

Correct answer: C

## EXAM SHORTCUT

Unbiasedness is immediate; independence across groups gives the variance formula (the Scheffé/Tukey denominators of S16).

Q12.

Consider the following statements about MSE in one-way ANOVA: I. MSE is the pooled within-group sample variance  $\sum_i (n_i - 1) s_i^2 / \sum_i (n_i - 1)$ . II.  $E(\text{MSE}) = \sigma^2$  whether or not  $H_0$  is true. Which of the statements given above is/are correct?

- A. Neither I nor II
- B. I only
- C. Both I and II
- D. II only

Correct answer: C

## EXAM SHORTCUT

MSE pools the within-group variances and is unbiased regardless of the hypothesis.

# Two-Way Classification: Models, Estimability, Degrees of Freedom

14 forecast questions in this topic

Linear Models

Two-Way Classification: Models, Estimability, Degrees of Freedom

Medium

**Q1.**

For a two-way classification with one observation per cell, factor A at 4 levels and factor B at 6 levels, the degrees of freedom for error are

A. 23

B. 24

**C. 45**

D. 18

**Correct answer: C**

## EXAM SHORTCUT

$(p-1)(q-1) = 3 \times 5 = 15$  (2023 Q61 archetype).

Linear Models

Two-Way Classification: Models, Estimability, Degrees of Freedom

Medium

**Q2.**

For a two-way classification with factor A at 3 levels, factor B at 4 levels and 5 observations per cell, the degrees of freedom for error are

A. 59

B. 44

**C. 48**

D. 60

**Correct answer: C**

## EXAM SHORTCUT

$pq(m-1) = 12 \times 4 = 48$  (2025 Q51 / 2022 Q7 archetype: there  $5 \cdot 6 \cdot 3 = 90$ ).

Linear Models

Two-Way Classification: Models, Estimability, Degrees of Freedom

Medium

**Q3.**

For a two-way ANOVA with 5 rows, 3 columns and 2 observations per cell, the degrees of freedom for interaction and error are respectively

**A. 8 and 15**

B. 8 and 14

C. 15 and 8

D. 10 and 15

**Correct answer: A**

## EXAM SHORTCUT

Interaction  $(5-1)(3-1) = 8$ ; error  $pq(m-1) = 15 \times 1 = 15$  (2018 Q6 archetype: there 15 and 48).

Q4.

In the model  $y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + e_{ijk}$  with  $i = 1, \dots, p$ ;  $j = 1, \dots, q$ ;  $k = 1, \dots, r$ , and  $p = 3$ ,  $q = 4$ , the total number of parameters (excluding  $\sigma^2$ ) is

- A. 12
- B. 20
- C. 8
- D. 19

Correct answer: B

## EXAM SHORTCUT

$1 + p + q + pq = 1 + 3 + 4 + 12 = 20$  (2018 Q15 archetype).

Q5.

For the additive two-way model  $y_{ij} = \mu + \alpha_i + \beta_j + e_{ij}$  (one observation per cell), the design matrix has  $1 + p + q$  columns. Its rank is

- A.  $p + q$
- B.  $1 + p + q$
- C.  $pq$
- D.  $p + q - 1$

Correct answer: D

## EXAM SHORTCUT

Two linear dependencies among the  $1 + p + q$  columns: rank  $p + q - 1$ .

Q6.

In the two-way model  $y_{ij} = \mu + \alpha_i + \beta_j + e_{ij}$ ,  $i = 1, \dots, p$ ;  $j = 1, \dots, q$ , which one of the following is estimable?

- A.  $\alpha_1 + \alpha_2$
- B.  $\alpha_i$
- C.  $\beta_j$
- D.  $\alpha_i - \alpha_u$  ( $i \neq u$ )

Correct answer: D

## EXAM SHORTCUT

2021 Q9 verbatim archetype: only differences of like effects (and cell expectations) are estimable.

**Q7.**

With one observation per cell in a two-way layout, consider the following statements: I. The interaction cannot be tested by the usual ANOVA because it is completely confounded with error. II. Tukey's one-degree-of-freedom test for non-additivity can nevertheless be used to detect a multiplicative interaction. Which of the statements given above is/are correct?

- A. II only
- B. I only
- C. Neither I nor II
- D. Both I and II

**Correct answer: D****EXAM SHORTCUT**

One observation per cell leaves no pure replicate error; Tukey's 1-df test rescues a specific form of non-additivity.

**Q8.**

In a two-way classification with one observation per cell, factor A at 4 levels and factor B at 6 levels, TSS = 120, SSA = 40 and SSB = 50. The mean square for error is

- A. 30
- B. 1.5
- C. 2
- D. 3

**Correct answer: C****EXAM SHORTCUT**

$SSE = 120 - 40 - 50 = 30$  on  $(4 - 1)(6 - 1) = 15$  df:  $MSE = 2$ .

**Q9.**

In the fixed-effects two-way model with  $m > 1$  observations per cell, consider the following statements: I. Each of  $H_0A$ ,  $H_0B$  and  $H_0(AB)$  is tested against MSE. II. The interaction hypothesis should be examined before interpreting the main-effect tests. Which of the statements given above is/are correct?

- A. Both I and II
- B. Neither I nor II
- C. II only
- D. I only

**Correct answer: A****EXAM SHORTCUT**

With replication all three ratios use MSE (fixed effects); interaction first is the standard protocol.

**Q10.**

Four methods are compared using six analysts, each analyst producing one observation under each method (two-way layout, one observation per cell). Given  $TSS = 150$ ,  $SS(\text{Analysts}) = 30$  and  $SS(\text{Methods}) = 84$ , the mean square error is

**A. 2.4****B. 15****C. 36****D. 3.0****Correct answer: A****EXAM SHORTCUT**

$SSE = 150 - 30 - 84 = 36$  on  $5 \times 3 = 15$  df:  $MSE = 2.4$  (2023 Q52 archetype: there  $13.73/8 = 1.72$ ).

**Q11.**

For a  $2 \times 2$  layout with 2 observations per cell, the degrees of freedom for A, B, interaction, error and total are respectively

**A. 1, 1, 2, 3, 7****B. 2, 2, 1, 2, 7****C. 1, 1, 1, 4, 7****D. 1, 1, 1, 3, 6****Correct answer: C****EXAM SHORTCUT**

1, 1, 1, error  $ab(m-1) = 4$ , total  $abm-1 = 7$ .

**Q12.**

Three teaching methods are tested with four instructors and 2 students per method–instructor cell. The degrees of freedom for methods, instructors, interaction and error are respectively

**A. 2, 3, 6, 13****B. 2, 3, 5, 13****C. 3, 4, 12, 24****D. 2, 3, 6, 12****Correct answer: D****EXAM SHORTCUT**

$(3-1), (4-1), (2)(3), 12 \cdot 1 = 12$  (2020 Q45 archetype).

**Q13.**

A randomized block design with  $b = 5$  blocks and  $v = 4$  treatments is analysed as a two-way classification with one observation per cell. The error degrees of freedom are

- A. 20
- B. 16
- C. 19
- D. 12**

**Correct answer: D**

**EXAM SHORTCUT**

$$(b-1)(v-1) = 4 \times 3 = 12.$$

**Q14.**

In the fixed-effects two-way model with one observation per cell, consider the following statements: I.  $E(\text{MSE}) = \sigma^2$  provided the additive model is correct. II. If interaction is actually present, MSE is inflated and the F tests for main effects lose power. Which of the statements given above is/are correct?

- A. I only
- B. II only
- C. Neither I nor II
- D. Both I and II**

**Correct answer: D**

**EXAM SHORTCUT**

Additivity is exactly what frees the  $(p-1)(q-1)$  cell for error; violated additivity biases MSE upward.

# Fixed, Random and Mixed Effects Models

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12 forecast questions in this topic

Linear Models

Fixed, Random and Mixed Effects Models

Medium

**Q1.**

Following Eisenhart's classification, the linear models with all effects fixed, all effects random, and some fixed with some random are called respectively

A. Model II, Model I and Model III

**B. Model I, Model II and Model III (mixed model)**

C. Model I, Model III and Model II

D. Model III, Model II and Model I

**Correct answer: B**

## EXAM SHORTCUT

Eisenhart's Model I (fixed), Model II (random), Model III (mixed).

Linear Models

Fixed, Random and Mixed Effects Models

Medium

**Q2.**

A factor should be treated as random when

A. it has exactly two levels

B. its levels are quantitative

C. its levels are the only levels of interest

**D. Its levels can be regarded as a random sample from a population of levels, and inference is sought for that population**

**Correct answer: D**

## EXAM SHORTCUT

Random = sampled levels + inference to the population of levels (the 2025 Q54 principle).

Linear Models

Fixed, Random and Mixed Effects Models

Medium

**Q3.**

In an RBD with  $v$  treatments and  $b$  blocks, the blocks used are regarded as a sample from a large population of blocks, while the treatments are the specific ones under study. The appropriate two-way model is

A. a fixed-effects model for both factors

B. a random-effects model for both factors

**C. A mixed model: treatments fixed, blocks random**

D. a mixed model: treatments random, blocks fixed

**Correct answer: C**

## EXAM SHORTCUT

2025 Q54 verbatim archetype: sampled blocks random, studied treatments fixed.

Q4.

In the balanced one-way random-effects model  $y_{ij} = \mu + a_i + e_{ij}$ ,  $a_i \sim (0, \sigma^2_a)$ ,  $e_{ij} \sim (0, \sigma^2)$ , all independent,  $j = 1, \dots, n$ , consider the following statements: I.  $E(\text{MS}_{\text{Between}}) = \sigma^2 + n\sigma^2_a$ . II.  $E(\text{MS}_{\text{Within}}) = \sigma^2$ . Which of the statements given above is/are correct?

- A. II only
- B. Both I and II**
- C. Neither I nor II
- D. I only

Correct answer: B

**EXAM SHORTCUT**

The two EMS lines that generate the F test and the variance-component estimator.

Q5.

In the model of F15.4 with  $n = 6$  observations per class,  $\text{MS}_{\text{Between}} = 50$  and  $\text{MS}_{\text{Within}} = 20$ . The ANOVA estimator of  $\sigma^2_a$  is

- A. 30
- B. 6
- C. 3.5
- D. 5**

Correct answer: D

**EXAM SHORTCUT**

$\hat{\sigma}^2_a = (\text{MSB} - \text{MSW})/n = (50 - 20)/6 = 5$ .

Q6.

Continuing F15.5 ( $\hat{\sigma}^2_a = 5, \hat{\sigma}^2 = 20$ ), the estimated intraclass correlation coefficient is

- A. 0.25
- B. 0.05
- C. 0.20**
- D. 0.80

Correct answer: C

**EXAM SHORTCUT**

$\hat{\rho} = \hat{\sigma}^2_a / (\hat{\sigma}^2_a + \hat{\sigma}^2) = 5/25 = 0.2$ .

Q7.

In the one-way random-effects model, consider the following statements: 1.  $V(y_{ij}) = \sigma^2_a + \sigma^2$ . 2.  $\text{Cov}(y_{ij}, y_{ij'}) = \sigma^2_a$  for  $j \neq j'$  (same class). 3. Observations from different classes are uncorrelated. Which of the above statements are correct?

A. 2 and 3 only

B. **1, 2 and 3**

C. 1 and 3 only

D. 1 and 2 only

Correct answer: B

## EXAM SHORTCUT

The random effect induces the intraclass covariance structure; classes are independent.

Q8.

In the balanced two-way random-effects model with  $m$  observations per cell (A at  $p$  levels, B at  $q$  levels), consider the following statements: I. MSE is the appropriate denominator for testing  $H_0(A \times B): \sigma^2_{ab} = 0$ . II. MSE is the appropriate denominator for testing  $H_0A: \sigma^2_a = 0$ . III.  $MS(AB)$  is the appropriate denominator for testing  $H_0A: \sigma^2_a = 0$  and  $H_0B: \sigma^2_b = 0$ . Which of the statements given above are correct?

A. **I and III only**

B. II and III only

C. I and II only

D. I, II and III

Correct answer: A

## EXAM SHORTCUT

$E(MSA) = \sigma^2 + m\sigma^2_{ab} + qm\sigma^2_a$  and  $E(MS(AB)) = \sigma^2 + m\sigma^2_{ab}$ : main effects are tested against  $MS(AB)$ , interaction against MSE (2024 Q59 archetype, keyed I only there because only its statement I was true).

Q9.

In the balanced two-way random-effects model with  $m$  observations per cell, the ANOVA estimator of  $\sigma^2_b$  is

A.  $(MSB - MS(AB))/(qm)$ B.  $(MSA - MSB)/(pq)$ C.  **$(MSB - MS(AB))/(pm)$** D.  $(MSB - MSE)/(qm)$ 

Correct answer: C

## EXAM SHORTCUT

Match EMS lines:  $E(MSB) - E(MS(AB)) = pm\sigma^2_b$  (2022 Q8 mirror image).

## Q10.

In the balanced two-way mixed model (A fixed, B random) with interaction and  $m$  observations per cell, the fixed factor A is tested by the ratio

- A.  $MSA/MSE$
- B.  $MSA/MS(AB)$
- C.  $MSA/MSB$
- D.  $MS(AB)/MSA$

Correct answer: B

## EXAM SHORTCUT

$E(MSA)$  contains  $m\sigma^2_{ab}$ ; the interaction mean square is the correct error term for the fixed factor in the mixed model.

## Q11.

Consider the following statements: I. In the fixed-effects one-way model the null hypothesis is  $H_0: \alpha_1 = \dots = \alpha_k$  (equivalently all  $\alpha_i = 0$  under the usual constraint). II. In the random-effects one-way model the corresponding null hypothesis is  $H_0: \sigma^2_a = 0$ . Which of the statements given above is/are correct?

- A. Both I and II
- B. I only
- C. Neither I nor II
- D. II only

Correct answer: A

## EXAM SHORTCUT

Same F statistic, different hypotheses — the conceptual heart of Model I vs. Model II.

## Q12.

Consider the following statements about ANOVA (method-of-moments) estimators of variance components: I. They can take negative values (e.g., when  $MS_{\text{Between}} < MS_{\text{Within}}$ ). II. The usual practical convention is to truncate a negative estimate to zero. Which of the statements given above is/are correct?

- A. Neither I nor II
- B. II only
- C. I only
- D. Both I and II

Correct answer: D

## EXAM SHORTCUT

Method-of-moments estimators are not constrained to the parameter space; truncation at 0 is standard.

# Multiple Comparison Tests: Tukey, Scheffé, SNK, Duncan

14 forecast questions in this topic

Linear Models

Multiple Comparison Tests: Tukey, Scheffé, SNK, Duncan

Medium

**Q1.**

Tukey's honestly-significant-difference (HSD) procedure is based on the distribution of

- A. the studentized range statistic  $q$**
- B. the sample correlation coefficient
- C. Snedecor's  $F$
- D. Student's  $t$

**Correct answer: A**

## EXAM SHORTCUT

HSD = studentized range; this single fact separates it from Scheffé ( $F$ -based) (2018 Q11/2021 Q6 family).

Linear Models

Multiple Comparison Tests: Tukey, Scheffé, SNK, Duncan

Medium

**Q2.**

Consider the following statements: 1. Scheffé's test of multiple comparisons tests all simple and complex contrasts. 2. Scheffé's procedure is derived from the  $F$  distribution. 3. Tukey's test of multiple comparisons reduces the Type-I error of the family of pairwise tests. Which of the above statements are correct?

- A. 2 and 3 only
- B. 1, 2 and 3**
- C. 1 and 2 only
- D. 1 and 3 only

**Correct answer: B**

## EXAM SHORTCUT

All three: Scheffé covers every contrast via the  $F$  distribution; Tukey controls the familywise Type-I error (the 2018 Q11 key, with the false Bartlett statement removed).

Linear Models

Multiple Comparison Tests: Tukey, Scheffé, SNK, Duncan

Medium

**Q3.**

Among Scheffé, Tukey, Student–Newman–Keuls and Duncan, when applied to all pairwise comparisons, the most conservative procedure (widest critical differences) is generally

- A. Student–Newman–Keuls
- B. Duncan
- C. Tukey
- D. Scheffé**

**Correct answer: D**

## EXAM SHORTCUT

For pairwise work Scheffé is widest (it pays for all contrasts); Duncan is the most liberal.

Q4.

In the Student–Newman–Keuls (SNK) procedure, the critical value used for comparing two ordered means depends on

- A. only the error degrees of freedom
- B. ~~A~~ the number of means spanned (the “stretch”  $p$ ) between the two ordered means
- C. only the total number of treatments  $k$
- D. the value of the grand mean

Correct answer: B

## EXAM SHORTCUT

Stepwise range test:  $qa; p, v$  with  $p$  the stretch of the two ordered means (2021 Q6 statement 2).

Q5.

Consider the following statements about Duncan’s multiple range test: I. It uses protection level  $(1 - \alpha)^{p-1}$  for a stretch of  $p$  means, so its critical values are smaller than those of SNK. II. It is more liberal (declares significance more readily) than Tukey’s HSD. Which of the statements given above is/are correct?

- A. II only
- B. Neither I nor II
- C. I only
- D. ~~B~~ Both I and II

Correct answer: D

## EXAM SHORTCUT

Duncan’s shrinking protection level makes it the high-power, low-protection member of the family.

Q6.

Three machines have mean outputs  $\bar{y}_A = 45$ ,  $\bar{y}_B = 52$ ,  $\bar{y}_C = 40$ , and the critical difference at the 5% level is 6.5. Which one of the following is correct?

- A. No two machines differ significantly
- B. Only machines B and C differ significantly
- C. ~~A~~ Machine B differs significantly from both A and C, while A and C do not differ
- D. Only machines A and B differ significantly

Correct answer: C

## EXAM SHORTCUT

$|A-B| = 7 > 6.5$ ,  $|B-C| = 12 > 6.5$ ,  $|A-C| = 5 < 6.5$  (2025 Q52 archetype: there only B vs. C crossed 5.62).

**Q7.**

Consider the following statements: I. For simple pairwise differences, Tukey's intervals are narrower than Scheffé's. II. For complex contrasts involving several means, Scheffé's method is the appropriate choice. Which of the statements given above is/are correct?

**A. ~~Both I and II~~****B.** II only**C.** Neither I nor II**D.** I only**Correct answer: A****EXAM SHORTCUT**

The standard division of labour between the two procedures.

**Q8.**

Consider the following statements about Scheffé's procedure: I. If the overall ANOVA  $F$  test is significant at level  $\alpha$ , at least one contrast is declared significant by Scheffé's criterion at the same level. II. If the overall  $F$  test is not significant, no contrast can be Scheffé-significant. Which of the statements given above is/are correct?

**A.** I only**B. ~~Both I and II~~****C.** II only**D.** Neither I nor II**Correct answer: B****EXAM SHORTCUT**

Scheffé is exactly compatible with the overall  $F$  test in both directions.

**Q9.**

With  $k = 5$  treatments, consider the following statements: I. There are 10 pairwise comparisons in all. II. Performing 10 separate  $t$ -tests each at level 0.05 inflates the familywise Type-I error well beyond 0.05. Which of the statements given above is/are correct?

**A. ~~Both I and II~~****B.** II only**C.** Neither I nor II**D.** I only**Correct answer: A****EXAM SHORTCUT**

$C(5,2) = 10$ ; familywise error  $\approx 1 - (0.95)^{10} \approx 0.40$  (2018 Q12 statement 1).

## Q10.

Consider the following statements: I. Tukey's HSD in its classical form requires equal sample sizes in all groups. II. The Tukey–Kramer modification extends the procedure to unequal sample sizes. Which of the statements given above is/are correct?

A. Neither I nor II

**B. Both I and II**

C. II only

D. I only

**Correct answer: B**

**EXAM SHORTCUT**

Equal  $n$  in the classical form; Tukey–Kramer handles unbalanced data.

## Q11.

With  $k$  treatments,  $n$  observations per treatment and error degrees of freedom  $v$ , Tukey's honestly significant difference is

**A.  $q_{\alpha; k, v} \sqrt{(MSE)/(n)}$**

B.  $t_{\alpha/2; v} \sqrt{2(MSE)/(n)}$

C.  $q_{\alpha; k, v} (MSE/n)$

D.  $\sqrt{(k-1)F_{\alpha; k-1, v}}$

**Correct answer: A**

**EXAM SHORTCUT**

$q$  already studentizes the range of  $k$  means, so the SE uses  $MSE/n$ , not  $2MSE/n$ .

## Q12.

For a contrast  $\psi = \sum c_i \bar{y}_i$  with standard error  $\hat{s}\psi = \sqrt{MSE \sum c_i^2 / n_i}$ , Scheffé's method declares  $\psi \neq 0$  at level  $\alpha$  when  $|\psi|$  exceeds

**A.  $\sqrt{(k-1)F_{\alpha; k-1, v}} \hat{s}\psi$**

B.  $F_{\alpha; k-1, v} \hat{s}\psi$

C.  $q_{\alpha; k, v} \hat{s}\psi$

D.  $t_{\alpha/2; v} \hat{s}\psi$

**Correct answer: A**

**EXAM SHORTCUT**

Scheffé's constant is  $\sqrt{(k-1)F_{\alpha}}$ —valid simultaneously for every contrast.

**Q13.**

Consider the following statements about Fisher's least significant difference (LSD): I. The "protected" LSD is applied only after the overall  $F$  test is significant. II. The LSD for comparing two equal-sized groups is  $t_{\alpha/2; \nu} \sqrt{2 \text{MSE}/n}$ . Which of the statements given above is/are correct?

- A. I only
- B. Neither I nor II
- C. **Both I and II**
- D. II only

**Correct answer: C**

**EXAM SHORTCUT**

Protected LSD = ordinary  $t$  yardstick, gated by the ANOVA  $F$ .

**Q14.**

Consider the following statements: I. Tukey's HSD controls the familywise error rate at  $\alpha$  for the family of all pairwise comparisons. II. Duncan's multiple range test does not control the familywise error rate at  $\alpha$ . Which of the statements given above is/are correct?

- A. I only
- B. II only
- C. **Both I and II**
- D. Neither I nor II

**Correct answer: C**

**EXAM SHORTCUT**

Tukey: exact FWER control (balanced case); Duncan trades protection for power (2018 Q12 statement 3 flavour).

# ANOVA: Concepts, Assumptions and Behaviour of F

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10 forecast questions in this topic

Linear Models

ANOVA: Concepts, Assumptions and Behaviour of F

Medium

## Q1.

Which of the following statements regarding the ANOVA technique are correct? I. It is used to test the equality of several population means. II. It is primarily a test of the equality of several population variances. III. Its origin lies in agricultural experiments. Select the answer using the code given below.

A. II and III only

B. I, II and III

C. **I and III only**

D. I and II only

**Correct answer: C**

### EXAM SHORTCUT

2024 Q57 verbatim archetype: ANOVA compares means (despite its name); Fisher's agricultural origin.

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Linear Models

ANOVA: Concepts, Assumptions and Behaviour of F

Medium

## Q2.

In ANOVA for one-way classified data, which of the following statements are correct? I. The sums of squares are unaffected by a change of origin. II. The sums of squares are unaffected by a change of scale. III. The variance ratio is unaffected by a change of origin. IV. The variance ratio is unaffected by a change of scale. Select the answer using the code given below.

A. **I, III and IV**

B. I, II and III

C. II, III and IV

D. I, II and IV

**Correct answer: A**

### EXAM SHORTCUT

2024 Q58 verbatim archetype: scale multiplies every SS by  $c^2$ , which cancels in the ratio.

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## Q3.

For the validity of the F test in ANOVA, which of the following are assumed? I. The observations are independent. II. The parent populations are normally distributed. III. The error variances are homogeneous (equal) across groups. IV. The various effects are additive in nature. Select the correct answer using the code given below.

- A. I, II and IV only
- B. I, II and III only
- C. I, III and IV only
- D. **I, II, III and IV**

**Correct answer: D**

**EXAM SHORTCUT**

2024 Q60 archetype: independence, normality, homoscedasticity, additivity — the trap option there asserted variances “may differ”.

## Q4.

As the variability due to chance (error variance) in an ANOVA decreases, other things remaining the same, the F statistic

- A. requires additional information to claim any change
- B. will remain constant
- C. **will always increase**
- D. will always decrease

**Correct answer: C**

**EXAM SHORTCUT**

2018 Q7 verbatim archetype: smaller MSE, larger  $F = MST/MSE$ .

## Q5.

When the  $k$  population means are truly different from each other, a significant F statistic is likely to arise because the average error deviation

- A. is relatively large compared to the average treatment deviations
- B. differs significantly between at least two of the populations
- C. **is relatively small compared to the average treatment deviations**
- D. equals the average treatment deviations

**Correct answer: C**

**EXAM SHORTCUT**

2018 Q8 verbatim archetype: signal large relative to noise.

Q6.

Consider the following statements: I. The ANOVA F test is a right-tailed (one-tailed) test. II. For  $k=2$  groups,  $F_{1,\nu} = t^2_{\nu}$ : one-way ANOVA is equivalent to the two-sample pooled  $t$  test. Which of the statements given above is/are correct?

- A. Neither I nor II
- B. I only
- C. **Both I and II**
- D. II only

Correct answer: C

#### EXAM SHORTCUT

Both classical facts; the square of  $t$  is  $F_{1,\nu}$ .

Q7.

Consider the following statements: I. The F test in balanced ANOVA is fairly robust to moderate departures from normality. II. Heterogeneity of variances is most damaging when the group sizes are unequal. Which of the statements given above is/are correct?

- A. I only
- B. **Both I and II**
- C. Neither I nor II
- D. II only

Correct answer: B

#### EXAM SHORTCUT

Standard robustness folklore, examinable as stated.

Q8.

To stabilize the variance of Poisson-type count data before ANOVA, the standard transformation is

- A. the arcsine transformation  $\sin^{-1} \sqrt{y}$
- B. no transformation is ever needed
- C. **the square-root transformation  $\sqrt{y}$**
- D. the reciprocal transformation  $1/y$

Correct answer: C

#### EXAM SHORTCUT

$\sqrt{y}$  for counts; arcsine for proportions; log when  $SD \propto \text{mean}$ .

**Q9.**

Consider the following statements: I. Bartlett's test is a test of homogeneity of variances, not a multiple-comparison test. II. Levene's test can be used for the same purpose and is less sensitive to non-normality. Which of the statements given above is/are correct?

- A. II only
- B. I only
- C. Neither I nor II
- D. **Both I and II**

**Correct answer: D****EXAM SHORTCUT**

The 2018 Q11 trap dressed correctly: Bartlett tests variances.

**Q10.**

If every observation in a one-way layout is transformed as  $z_{ij} = cy_{ij} + d$  with  $c \neq 0$ , consider the following statements: I. Every sum of squares is multiplied by  $c^2$ . II. The F ratio computed from the  $z$ 's equals the F ratio computed from the  $y$ 's. Which of the statements given above is/are correct?

- A. II only
- B. I only
- C. Neither I nor II
- D. **Both I and II**

**Correct answer: D****EXAM SHORTCUT**

Origin shift cancels within deviations; scale cancels between numerator and denominator.

# Numerical ANOVA Drills: Complete the Table

11 forecast questions in this topic

Linear Models

Numerical ANOVA Drills: Complete the Table

Medium

**Q1.**

In a one-way ANOVA the error degrees of freedom are 15, the total degrees of freedom 17,  $MSE = 10$  and the total sum of squares is 500. The values of (SS Treatment, SS Error, MS Treatment, F) are

- A. (150, 350, 75, 7.5)
- B. (350, 150, 116.7, 11.67)
- C. (300, 200, 150, 15)
- D. **(350, 150, 175, 17.5)**

**Correct answer: D**

## EXAM SHORTCUT

$SSE = 15 \times 10 = 150$ ;  $SSTr = 500 - 150 = 350$  on  $17 - 15 = 2$  df;  $MST = 175$ ;  $F = 17.5$  (2019 Q5 / 2023 Q65 archetype).

Linear Models

Numerical ANOVA Drills: Complete the Table

Medium

**Q2.**

In the following two-way ANOVA table (one observation per cell) a few entries are missing: Source d.f. SS Variance ratio Treatment 24.6 4.1 Variety  $x$  7.5 Error  $y$   $z$  Total The values of  $x$ ,  $y$  and  $z$  are respectively

- A. **2, 6, 12**
- B. 3, 5, 10
- C. 2, 6, 15
- D. 2, 5, 12

**Correct answer: A**

## EXAM SHORTCUT

$MST = 24.6/3 = 8.2$  and ratio 4.1  $\Rightarrow MSE = 2$ ;  $7.5 = (30/x)/2 \Rightarrow x = 2$ ;  $y = 11 - 3 - 2 = 6$ ;  $z = 6 \times 2 = 12$  (2025 Q56 archetype).

Linear Models

Numerical ANOVA Drills: Complete the Table

Medium

**Q3.**

A one-way layout has  $k = 4$  treatments and  $N = 24$  observations in all, with  $TSS = 600$  and treatment sum of squares = 300. The F statistic equals

- A. 10.00
- B. 4.44
- C. 5.00
- D. **6.67**

**Correct answer: D**

## EXAM SHORTCUT

$MST = 300/3 = 100$ ;  $MSE = 300/20 = 15$ ;  $F = 100/15 = 6.67$ .

**Q4.**

Four methods are compared by six analysts (two-way, one observation per cell). Given  $TSS = 100$ ,  $SS(\text{Analysts}) = 10$ ,  $SS(\text{Methods}) = 60$ , the MSE is

**A. 2.0**

B. 1.5

C. 3.0

D. 2.4

**Correct answer: A****EXAM SHORTCUT**

$SSE = 100 - 10 - 60 = 30$  on  $5 \times 3 = 15$  df:  $MSE = 2$  (2023 Q52 archetype).

**Q5.**

A two-way experiment has factor A at 4 levels, factor B at 3 levels and 3 observations per cell. The degrees of freedom for A, B, interaction and error are respectively

**A. 3, 2, 6, 24**

B. 3, 2, 5, 25

C. 3, 2, 6, 23

D. 4, 3, 12, 24

**Correct answer: A****EXAM SHORTCUT**

3, 2, 6, error =  $pq(m-1) = 12 \times 2 = 24$  (2020 Q45 archetype).

**Q6.**

In a two-way layout with factor A at 2 levels, factor B at 3 levels and 2 observations per cell, the sums of squares are  $SSA = 0.5$ ,  $SSB = 24$ ,  $SSE = 0.66$ . The mean squares for A and error are respectively

A. 0.50 and 0.132

**B. 0.50 and 0.11**

C. 0.25 and 0.11

D. 0.25 and 0.066

**Correct answer: B****EXAM SHORTCUT**

d.f.: A = 1, error =  $ab(m-1) = 6$ :  $MSA = 0.5$ ,  $MSE = 0.11$  (2022 Q9 archetype).

**Q7.**

Continuing F18.6, the F statistics for factors A and B are respectively (to two decimals)

- A. 2.27 and 109.09
- B. **4.55 and 109.09**
- C. 4.55 and 218.18
- D. 0.76 and 36.36

**Correct answer: B****EXAM SHORTCUT**

$MSB = 24/2 = 12$ ;  $FA = 0.5/0.11 = 4.55$ ,  $FB = 12/0.11 = 109.09$  (2022 Q10 archetype).

**Q8.**

Fish are sampled from three lakes, five observations from each lake ( $N = 15$ ). The total sum of squares of all 15 observations is 350. If all the observations are pooled into a single sample, their standard deviation equals

- A. 4.83
- B. 25.00
- C. **5.00**
- D. 5.92

**Correct answer: C****EXAM SHORTCUT**

Pooled variance =  $TSS/(N-1) = 350/14 = 25$ ;  $SD = 5$  (2020 Q44 archetype).

**Q9.**

In a balanced fixed-effects one-way model with  $k = 3$ ,  $n = 5$  observations per group,  $(\alpha_1, \alpha_2, \alpha_3) = (-2, 0, 2)$  and  $\sigma^2 = 4$ , the expected value of the treatment mean square is

- A. 44
- B. **24**
- C. 20
- D. 4

**Correct answer: B****EXAM SHORTCUT**

$E(MST) = \sigma^2 + n \sum \alpha_i^2 / (k-1) = 4 + 5(8)/2 = 24$ .

**Q10.**

In a balanced two-way random-effects model with  $p=3$ ,  $q=4$ ,  $m=2$ , the mean squares are  $MSA = 60$  and  $MS(AB) = 20$ . The ANOVA estimate of  $\sigma^2_a$  is

**A. 5**

B. 6.67

C. 10

D. 40

**Correct answer: A****EXAM SHORTCUT**
$$\sigma^2_a = (MSA - MS(AB))/(qm) = (60-20)/8 = 5 \text{ (2022 Q8 archetype).}$$
**Q11.**

In a balanced one-way random-effects model with  $n=4$  observations per class,  $MS_{\text{Between}} = 34$  and  $MS_{\text{Within}} = 10$ . The estimates of  $\sigma^2_a$  and of the intraclass correlation are respectively

A. 6 and 0.600

**B. 6 and 0.375**

C. 24 and 0.706

D. 8 and 0.444

**Correct answer: B****EXAM SHORTCUT**
$$\sigma^2_a = (34-10)/4 = 6; \hat{\rho} = 6/(6+10) = 0.375.$$